Precision and Decisiveness as Goals: Reliable Sequential Hypothesis Testing with a Dual Stopping Criterion

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Sequential hypothesis testing offers flexibility over fixed-sample designs, but stopping rules coupled to decision criteria risk confirmation bias through early peeking. The HDI+ROPE algorithm exemplifies this: it stops as soon as the posterior Highest Density Interval (HDI) falls entirely outside a Region of Practical Equivalence (ROPE) around the null (rejecting it) or entirely inside, which enables positive acceptance of the null, something Null Hypothesis Significance Testing cannot do. However, this coupling means a wide HDI can satisfy the criterion on early, unrepresentative samples, incurring a systematic false-positive cost. Fully decoupled methods, such as "Precision is the Goal" (PitG), eliminate this bias by halting only once a target HDI width ω_{goal} is reached; yet, by divorcing the stopping rule from the decision criterion, PitG frequently yields inconclusive outcomes, particularly when the null hypothesis is true. We propose "Decisive Precision is the Goal" (DPitG), which requires both the precision target and a conclusive verdict to be satisfied simultaneously. In fair coin simulations ($\omega_{\text{goal}} = 0.08$, ROPE$= 0.5 \pm 0.05$), DPitG reduces the PitG inconclusive rate from 62% to 2% at a median cost of only 5% more samples, with zero false positives; HDI+ROPE achieves comparable conclusiveness only at the cost of false positives. Across the ω_{goal} range tested, DPitG conclusiveness remains above 97% while PitG's varies widely. We provide a closed-form planning formula $N_{\text{goal}} \propto V/\omega^2_{\text{goal}}$ (where V is the observation variance), an online interactive calculator, and open-source code. Demonstrated on single-group binary data, the framework extends readily to continuous outcomes and two-group comparisons. DPitG's fully pre-specified stopping rule is compatible with pre-registration standards and is the method of choice whenever a reliable verdict is required provided the study can afford to collect at least N_{goal} observations.

1 Introduction

Empirical hypothesis testing is foundational to scientific progress, enabling researchers to assess how well a sample represents an underlying truth. A central challenge in this process is determining when enough data have been collected to make a reliable decision, especially as the accuracy of statistical assessments depends heavily on sample size. Careful planning for statistical power and precision is essential to avoid biased inference and ensure meaningful results (Cohen, 1988).

Sequential hypothesis testing offers a flexible alternative to fixed-sample designs (Wald, 1947) by allowing data to be evaluated as it accumulates. This approach is widely used in fields such as clinical trials (Jennison and Turnbull, 2000; U.S. Food and Drug Administration, 2010), quality control, A/B testing (Johari et al., 2022), and financial monitoring, offering potential benefits in efficiency, timeliness, and ethical considerations. Central to this approach is the distinction between two critical components. The *stopping rule* dictates when data collection ends; the *decision rule* then determines the final verdict on the hypothesis, one of three outcomes: accept, reject, or inconclusive. While these components are conceptually distinct, many operational methods conflate them, triggering a stop exactly when a decision criterion is met.

This simultaneous application of stopping and decision rules is common in widely used frequentist and Bayesian methods, such as p-value thresholds (Fisher, 1925) or posterior-based heuristics (e.g., HDI+ROPE Kruschke, 2011, 2013, 2015, Bayes Factors Kass and Raftery, 1995). However, if not carefully designed, this coupling can lead to confirmation bias: premature stopping on extreme or unrepresentative early samples, a phenomenon known as early peeking (Simmons et al., 2011). As a concrete illustration: in the fair coin setting ($\theta_{\text{true}} = \theta_{\text{null}} = 0.5$, credible interval of 95% and minimum effect size ± 0.05), a sequentially monitored HDI+ROPE algorithm achieves high conclusiveness but at a systematic 6.3% false-positive rate (Section 3.1).

One principled remedy is to decouple the stopping criterion entirely from the hypothesis outcome, halting based on posterior precision rather than the direction of the result.

Following the *Accuracy in Parameter Estimation* paradigm (Maxwell et al., 2008), Kruschke (2015) advocated for a method that strictly decouples these two steps, thereby addressing the risk of premature and biased conclusions from early peeking. He proposed using a predetermined target for posterior precision as the sole *stopping rule*, independent of the hypothesis outcome. Only once this precision goal is met is the *decision rule* applied to accept or reject the null hypothesis (via a posterior credible interval criterion). Kruschke (2015) demonstrated that this "Precision is the Goal" (PitG) method eliminates the early-peeking bias in outcome sampling. However, our

analysis reveals an important limitation: when the null hypothesis is true, or practically so, PitG often yields a high rate of inconclusive results, rather than correctly accepting the null.[1]

In this paper, we propose an improved variant we call "Decisive Precision is the Goal" (DPitG), in which precision is a necessary but no longer sufficient stopping condition: data collection continues until *both* the precision goal and a decisive hypothesis outcome are satisfied simultaneously. The benefit is most pronounced precisely where PitG's inconclusive rate is highest, namely when the null is true or practically so, making DPitG particularly valuable for confirming the null hypothesis. While scientific incentives often favour the discovery of new effects, rigorous confirmation of the null is equally critical in many fields: from bioequivalence trials for generic drugs (Schuirmann, 1987; Karalis and Macheras, 2012) to do-no-harm regression testing in software deployment and futility stopping in programme evaluations. Unlike *Null Hypothesis Significance Testing* (NHST, see Appendix B), which can only fail to reject the null, the precision-based framework enables positive acceptance as a genuine and actionable verdict, and DPitG ensures this verdict is reliably reached. This fully pre-specified rule is also compatible with pre-registration standards (Nosek et al., 2018) and directly addresses the design-stage imprecision implicated in the replication crisis (Open Science Collaboration, 2015; Gelman and Carlin, 2014).

We compare DPitG and PitG against the HDI+ROPE algorithm, a coupled baseline that stops as soon as the posterior yields a decisive result, through simulations with single-group dichotomous data.[2] Applied to this setting, DPitG substantially outperforms both alternatives: for example, at $\omega_{\text{goal}} = 0.08$, it reduces PitG's 62.1% inconclusive rate to 2.2% at a median cost of only 4.7% extra samples, while preserving a zero false-positive rate, an advantage the HDI+ROPE algorithm matches only at a 6.2% false-positive cost (Section 3.1). These advantages hold broadly across precision goals and true parameter values (Sections 3.1–3.2). The framework extends to continuous outcomes (Appendix C) and two-group comparisons (Appendix D). Alongside the empirical results we provide a closed-form planning formula, an online interactive calculator, and open-source code to support adoption in applied settings.

Section 2 describes the theoretical framework and the three stopping algorithms; Section 3 presents simulation results; and Section 4 places the findings in methodological context.

2 Methods

We study three algorithms built on two summary properties of the posterior distribution: its *location* (how far the estimate sits from the null, conveying the effect size) and its *width* (how uncertain that estimate is)[3]. Location is formalised by where the posterior credible interval sits relative to a pre-specified equivalence region around the null; width by how broad that interval is. Together they inform a shared Decision Rule that all three algorithms apply once their respective stopping criteria are met.

Section 2.1 introduces the shared inference components, namely the ROPE, the HDI, and the Decision Rule, that all three algorithms apply identically. Section 2.2 describes the three algorithms, which differ only in their stopping criteria. Section 2.3 derives the expected stopping iteration for precision-based methods analytically. Section 2.4 details the simulation design used to evaluate and compare all three algorithms.

2.1 Shared Components

2.1.1 Effect Size by Region of Practical Equivalence (ROPE)

A common misconception in hypothesis testing is that a "statistically significant" outcome is sufficient for real-world decision-making. In practice, one must consider the *effect size* (Cohen, 1988): how large is the difference, not merely whether it is detectable. For example, medical research requires a *minimal clinically important difference* (MCID), the smallest change that a patient or clinician would consider meaningful, for an outcome to justify a change in treatment (Jaeschke et al., 1989).

Consider a hypothetical scenario where a researcher examines whether a therapeutic differs in impact by gender. Even if the data yield a "statistically significant" result (say, the drug is 72.1% beneficial for males and 72.3% for females), a practitioner may consider this equivalent for all practical purposes. Hence, the researcher should define in advance a pre-specified equivalence margin around the null hypothesis.

One principled way to operationalise effect size is the *Region of Practical Equivalence* (ROPE; Kruschke 2011). The ROPE is defined as an area around the null value considered "similar enough to the null hypothesis" such that if the true value is within this area, it is effectively the same as the null hypothesis.

[1] Kruschke 2015 noted in passing that PitG "did remain undecided in some cases" (§13.3.2, page 391); our analysis shows this is a systematic and striking effect, with the sampling budget $N_{\max}$ and precision goal ω_{goal} as the key governing parameters. See below for quantitative examples.

[2] Two other widely used approaches, NHST (p-value-based) and Bayes Factors, are not among the three algorithms studied here. Both lack an explicit effect-size criterion (the ROPE); NHST cannot formally accept the null, while Bayes Factors lack a direct precision goal. Their limitations are illustrated in Appendices B and F, respectively.

[3] Here *epistemic uncertainty* means the uncertainty in our knowledge about the true value of a parameter, as opposed to *aleatory uncertainty*, which is the inherent variability of the system being studied.

The appropriate ROPE width is domain-specific: a board game manufacturer requires dice to be fair to a reasonable standard, but not to a precision indistinguishable to a casual player; conversely, a casino requires a much stricter tolerance (a narrower ROPE) to ensure regulatory compliance and fair play. Appendix G discusses principled approaches for setting the ROPE, including domain standards such as the MCID and bioequivalence guidelines, and robustness checks across a range of ROPE widths when no domain standard exists.

2.1.2 Highest Density Interval (HDI)

To quantify the uncertainty in our estimate we use the *credible interval*, the Bayesian analogue of the frequentist confidence interval: the region of the parameter space that contains a specified fraction of the posterior mass (e.g. 95%). Several rules exist for selecting which region to report; the two most common are the *Equal-Tailed Interval* (ETI) and the *Highest Density Interval* (HDI) (see Appendix A for their formal definitions).

The HDI has a natural geometric interpretation: imagine drawing a horizontal line across the posterior density curve and slowly lowering it from the peak. The line intersects the curve at two points, enclosing an area underneath. The HDI is the interval between those two intersection points at the moment the enclosed area first reaches the target mass (e.g. 95%). Because the line is horizontal, every point inside the interval sits on a portion of the curve that is at least as tall as any point outside — hence *highest density*. For a symmetric, bell-shaped posterior this coincides with the ETI (equal tails), but for a skewed posterior the HDI is noticeably shorter.

Throughout this work we use the HDI at the 95% level to remain comparable with traditional frequentist hypothesis testing.[4]

2.1.3 The Decision Rule

We distinguish between the *Stopping Rule* (when to stop collecting data) and the *Decision Rule* (what to conclude once stopped). While the three algorithms discussed later differ in their *stopping* criteria, they all share an identical Decision Rule.

The decision is determined solely by the relationship between the HDI and the ROPE. Let $\text{ROPE} = [\text{ROPE}_{\text{min}}, \text{ROPE}_{\text{max}}]$ and the posterior HDI limits be $[\text{HDI}_{\text{min}}, \text{HDI}_{\text{max}}]$, we define three outcomes based on their relative positions:

(a) *Accept Null*: The HDI is completely *inside* the ROPE.
$(\text{ROPE}_{\text{min}} \leq \text{HDI}_{\text{min}}) \wedge (\text{HDI}_{\text{max}} \leq \text{ROPE}_{\text{max}})$

(b) *Reject Null*: The HDI is completely *outside* the ROPE.
$(\text{HDI}_{\text{max}} < \text{ROPE}_{\text{min}}) \vee (\text{ROPE}_{\text{max}} < \text{HDI}_{\text{min}})$

(c) *Inconclusive*: The HDI *straddles* the ROPE boundary.
Otherwise.

The interpretation of these outcomes, particularly what it means to *accept* the null in a Bayesian framework, differs meaningfully from the frequentist convention and is worth examining carefully; we do so in Section 2.1.4.

Figure 1 illustrates all three outcomes using a hand-picked Bernoulli experiment ($\theta_{\text{null}} = 0.5$, $\text{ROPE}_{\text{min}}^{\text{max}} = [0.45, 0.55]$). In the top panel, a stop at iteration 126 finds the 95% HDI fully outside the ROPE, leading to rejection of the null hypothesis. In the middle panel, a stop at iteration 598 finds the HDI straddling the ROPE boundary, resulting in an inconclusive decision. In the bottom panel, a stop at iteration 804 finds the 95% HDI fully within the ROPE, allowing us to accept θ_{null}.

Figure 1 illustrates that the decision outcome may be iteration-dependent. An inconclusive outcome implies that the data collected is insufficient to justify a decision. In a live setting, a human-in-the-loop (or an algorithm, common in automated high-frequency environments) would need to either conduct a risk assessment or collect more data. Since risk assessment is highly contextual, we defer that discussion and focus on the necessity of collecting more data to avoid inconclusive outcomes.

Algorithm 1 further formalises this logic.

2.1.4 Interpreting Accept and Reject Decisions

A key conceptual distinction between Bayesian and frequentist inference lies in what it means to *accept* the null hypothesis. In NHST, a small p-value justifies *rejecting* the null; a large p-value only means there is insufficient evidence to reject it. One cannot formally *accept* the null, only "fail to reject" it. Absence of evidence is not evidence of absence: NHST can conclusively reject the null but cannot deliver its symmetric counterpart, a positive verdict that the null actually holds.

[4] Kruschke (2015) notes that for numerical posteriors obtained via Markov Chain Monte Carlo (MCMC), 95% HDI estimation requires at least 10,000 samples; a slightly lower level such as 94% is recommended for fewer samples. McElreath (2016) advocates 89% (the highest prime below 100, to underscore the arbitrariness of any threshold). Because we employ analytical posteriors in our experiments (Section 2.4), sampling noise is not a concern and 95% is unambiguous; practitioners relying on sampled (e.g., MCMC-based) posteriors should consider the lower thresholds noted above.

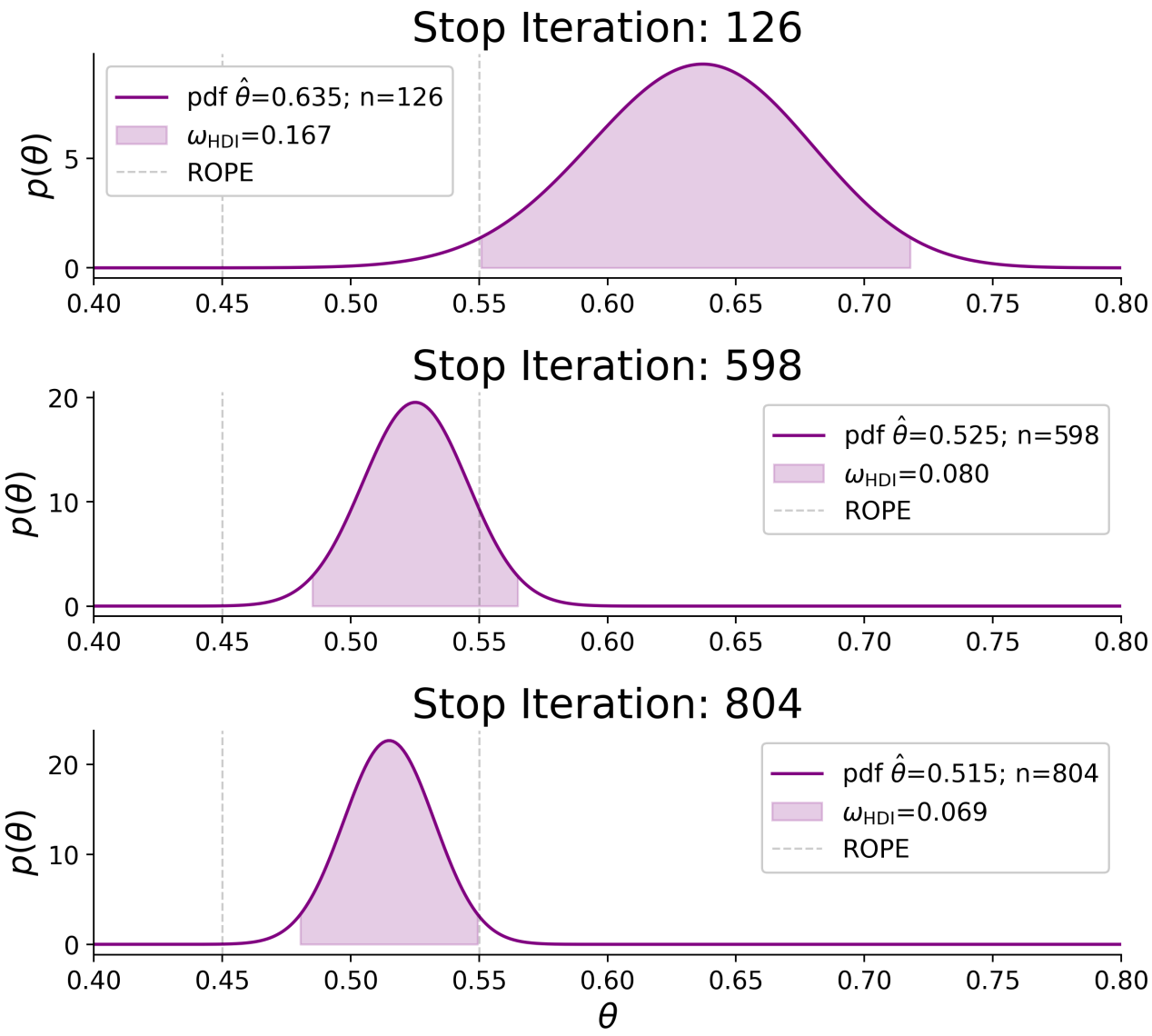


Figure 1: Posteriors at three stop iterations from a hand-picked experiment illustrating the three possible decision outcomes. Shaded areas are 95% HDIs; their widths ω_{HDI} are annotated. Vertical dashed lines mark the ROPE boundaries. *Top* (iteration 126): HDI fully outside the ROPE → Reject θ_{null}. *Middle* (iteration 598): HDI straddles the ROPE → Inconclusive. *Bottom* (iteration 804): HDI fully within the ROPE → Accept θ_{null}. See Figure 3 for the full sequence of iterations for this experiment.

The HDI+ROPE framework resolves this asymmetry: because the ROPE encodes a pre-specified notion of practical equivalence, each verdict carries positive evidential weight: Accept and Reject are both affirmative findings, not merely the presence or absence of a signal.[5] The HDI+ROPE framework's Accept verdict is therefore distinctive in being grounded in a pre-specified effect-size criterion.

Neither Accept nor Reject identifies the sample estimate $\hat{\theta}$ as the true value; both are statements about θ_{null}, not $\hat{\theta}$. The posterior distributes probability continuously across parameter space, and the HDI+ROPE framework distils this into a verdict by asking whether the bulk of posterior mass lies inside, outside, or straddling the ROPE. The full posterior remains available for a more nuanced reading: in particular, a "Reject null" decision does not endorse $\hat{\theta}$ as the true value; $\hat{\theta}$ is the mode (or mean) of the posterior, and the full distribution quantifies the uncertainty surrounding it.

The panels of Figure 1 make this concrete under the default setup ($\theta_{\text{null}} = 0.5$, $\text{ROPE}^{\text{min}}_{\text{max}} = [0.45, 0.55]$):

(a) Bottom panel (iteration 804, Accept): $\hat{\theta} = 0.515$ and the 95% HDI is [0.480, 0.549]. Both bounds lie inside the ROPE, so we accept $\theta_{\text{null}} = 0.5$: positive evidence that the true rate is practically equivalent to 0.5. We

[5]Note that Bayes Factors can also yield positive evidence for the null, but do so by comparing marginal likelihoods rather than by assessing practical equivalence (see Appendix F).

Algorithm 1 The Decision Rule

Require: ROPE_{min}, ROPE_{max}, HDI_{min}, HDI_{max}
  if ($\text{ROPE}_{\text{min}} \leq \text{HDI}_{\text{min}}$) & ($\text{HDI}_{\text{max}} \leq \text{ROPE}_{\text{max}}$) **then**
    Decision = Accept θ_{null} ▷ HDI completely within ROPE
  else if $\text{ROPE}_{\text{max}} < \text{HDI}_{\text{min}}$ **then**
    Decision = Reject θ_{null} ▷ HDI completely outside ROPE; $\theta_{\text{null}} < \hat{\theta}$
  else if $\text{HDI}_{\text{max}} < \text{ROPE}_{\text{min}}$ **then**
    Decision = Reject θ_{null} ▷ HDI completely outside ROPE; $\hat{\theta} < \theta_{\text{null}}$
  else
    Decision = Inconclusive ▷ HDI straddles ROPE
  end if
  return Decision

are *not* claiming the true rate equals 0.515.

(b) Top panel (iteration 126, Reject): $\hat{\theta} = 0.635$ and the 95% HDI is [0.551, 0.718]. Both bounds lie above $\text{ROPE}_{\text{max}} = 0.55$, so we reject $\theta_{\text{null}} = 0.5$: positive evidence that the true rate is meaningfully above 0.5. We are *not* claiming the true rate equals 0.635.

Two error types follow directly. A *false-accept* arises when a conclusive Accept verdict is reached but θ_{true} lies outside the ROPE, the HDI+ROPE framework analogue of a false negative, since the data appear to confirm equivalence when it does not hold. A *false-reject* arises when a conclusive Reject verdict is reached but θ_{true} lies inside the ROPE, the analogue of a false positive, since the data appear to refute a null that is in fact practically true. Their rates across θ_{true} values are characterised in Section 3.

2.2 Sequential Stopping Algorithms

Table 1 provides a high-level comparison of the three algorithms. Each algorithm shares the same Decision Rule (Algorithm 1) but differs in when it triggers the stop condition. Because the term "HDI+ROPE" serves two distinct roles in this work, we use distinct names throughout. *HDI+ROPE framework* denotes the shared Decision Rule applied identically by all three algorithms (Section 2.1.3). *HDI+ROPE algorithm* (equivalently, *HDI+ROPE stopping rule*) denotes the specific sequential stopping algorithm described in Section 2.2.1.

Algorithm	**Key Input**	**Stop Criterion**	**Precision/Location Coupling**
HDI+ROPE	N_{min}	Decision is conclusive	Location only
Precision is the Goal	ω_{Goal}	$\omega_{\text{HDI}} \leq \omega_{\text{Goal}}$	Precision then Location
Decisive PitG	ω_{Goal}	$\omega_{\text{HDI}} \leq \omega_{\text{Goal}}$ AND Decision is conclusive	Precision + Location

Table 1: Comparison of stopping algorithms. 'Decision is conclusive' means the HDI is fully inside or fully outside the ROPE. ω_{HDI} denotes the observed width of the HDI and ω_{Goal} denotes the target precision width.

The Key Input column identifies the characteristic parameter that must be determined before data collection: N_{min} for the HDI+ROPE algorithm (sensitive to early outliers and therefore requiring a minimum sample size, e.g. 30) and ω_{Goal} for the precision methods, where by definition ω_{Goal} must be narrower than or equal to the ROPE width Δ_{ROPE} to allow the HDI to fit fully within the ROPE.

The table also excludes two universal parameters required by all three algorithms:

(a) The ROPE boundaries ROPE_{min} and ROPE_{max}, with $\Delta_{\text{ROPE}} \equiv \text{ROPE}_{\text{max}} - \text{ROPE}_{\text{min}}$.

(b) N_{max} (Final Budget): The maximum sample size or budget limit.

All three algorithms below assume one data point is collected per iteration (Algorithms 2–4); the increment step (N += 1) can be replaced by a batch size to accommodate grouped collection.

2.2.1 HDI+ROPE Stopping: Location Only

The HDI+ROPE algorithm stops data collection as soon as the posterior's location allows for a decisive acceptance or rejection of the null hypothesis. It does not explicitly demand a specific degree of precision (width), only that the HDI does not straddle the ROPE boundary.

Algorithm 2 HDI+ROPE algorithm (highlighted: the stopping condition)

Require: N_{min}, N_{max}, θ_{null}, ROPE_{min}, ROPE_{max}
  Stop = False
  Decision = Inconclusive
  N = 0
  while (Stop = False & $N < N_{\text{max}}$) **do**
    N += 1 ▷ collect another data point
    Update $\hat{\theta}$, $P(\theta|\hat{\theta})$ ▷ update posterior
    HDI_{min}, $\text{HDI}_{\text{max}} \leftarrow P(\theta|\hat{\theta})$
    if $N \geq N_{\text{min}}$ **then**
      Decision = Decision Algorithm($\text{ROPE}_{\text{min}}, \text{ROPE}_{\text{max}}, \text{HDI}_{\text{min}}, \text{HDI}_{\text{max}}$)
      if Decision in {Accept θ_{null}, Reject θ_{null}} **then**
        Stop = True ▷ Stopping depends on decisive Decision
      end if
    end if
  end while

Algorithm 2 details the procedure. Note the inclusion of N_{min} (minimal sample size); this is critical because, with small sample sizes, the HDI can fluctuate wildly and accidentally fall entirely within or outside the ROPE due to random noise (aleatoric uncertainty)[6] rather than true effect.

This method serves as the coupled baseline throughout: lacking a precision requirement, it is susceptible to false-reject and false-accept decisions, as demonstrated in Section 3.

2.2.2 Precision is the Goal Stopping: Precision then Decision

Introduced by Kruschke (2015), Precision is the Goal shifts the stopping condition from the *decision* to the *precision* (width) of the estimate. Data collection stops when the width of the HDI is narrower than a pre-specified Goal (e.g., $\omega_{\text{Goal}} = 0.8\Delta_{\text{ROPE}}$).

Algorithm 3 shows that this method separates stopping from decision: stopping depends on width alone, while the decision depends on location. As we later demonstrate in Section 3, this can lead to situations where the algorithm stops (precision reached) but the result is inconclusive (posterior straddles the ROPE).

Algorithm 3 Precision is the Goal (highlighted: the stopping condition)

Require: ω_{Goal}, N_{max}, θ_{null}, ROPE_{min}, ROPE_{max}
  Stop = False
  Decision = Inconclusive
  N = 0
  while (Stop = False & $N < N_{\text{max}}$) **do**
    N += 1 ▷ collect another data point
    Update $\hat{\theta}, P(\theta|\hat{\theta})$ ▷ update posterior
    $\text{HDI}_{\text{min}}, \text{HDI}_{\text{max}} \leftarrow P(\theta|\hat{\theta})$
    $\omega_{\text{HDI}} = \text{HDI}_{\text{max}} - \text{HDI}_{\text{min}}$
    if $\omega_{\text{HDI}} \leq \omega_{\text{Goal}}$ **then**
      Stop = True ▷ Stopping depends on Precision ω_{HDI} regardless of Decision
      Decision = Decision Algorithm($\text{ROPE}_{\text{min}}, \text{ROPE}_{\text{max}}, \text{HDI}_{\text{min}}, \text{HDI}_{\text{max}}$)
    end if
  end while

2.2.3 Decisive Precision is the Goal Stopping: Precision and Decision

We propose Decisive Precision is the Goal, a conservative variant of PitG, to mitigate the high rate of inconclusive results. Unlike PitG, this method joins stopping and decision: it stops strictly when *both* the precision goal is met *AND* a decisive decision can be made.

As seen in Algorithm 4, strictly enforcing both conditions ensures that we do not stop in an indecisive state, potentially at the cost of collecting more data (up to N_{max}).

[6] Note the distinction from the epistemic uncertainty discussed earlier: the HDI *width* ω_{HDI} is epistemic: it reflects how much we do not yet know about θ and shrinks as N grows. The HDI *location*, however, varies across experiments due to the randomness of which particular draws are observed; at small N this aleatoric variability dominates, and the HDI can wander into a misleadingly decisive position.

Algorithm 4 Decisive Precision is the Goal (highlighted: the stopping condition)

Require: ω_{Goal}, $N_{\max}$, θ_{null}, $\text{ROPE}_{\min}$, $\text{ROPE}_{\max}$
  Stop = False
  Decision = Inconclusive
  N = 0
  while (Stop = False & $N < N_{\max}$) **do**
    N += 1 ▷ collect another data point
    Update $\hat{\theta}, P(\theta|\hat{\theta})$ ▷ update posterior
    $\text{HDI}_{\min}$, $\text{HDI}_{\max} \leftarrow P(\theta|\hat{\theta})$
    $\omega_{\text{HDI}} = \text{HDI}_{\max} - \text{HDI}_{\min}$
    if $\omega_{\text{HDI}} \leq \omega_{\text{Goal}}$ **then**
      Decision = Decision Algorithm($\text{ROPE}_{\min}, \text{ROPE}_{\max}, \text{HDI}_{\min}, \text{HDI}_{\max}$)
      if Decision in {Accept θ_{null}, Reject θ_{null}} **then**
        Stop = True ▷ Stopping depends on Precision ω_{HDI} and Decision
      end if
    end if
  end while

2.3 Expected Precision-Based Stop Iteration

For models where the posterior concentrates around the true parameter as data accumulate, a property of all regular parametric families, guaranteed asymptotically by the Bernstein–von Mises theorem (van der Vaart, 1998), the HDI width shrinks proportionally to $N^{-1/2}$. This makes the expected minimum stopping iteration of precision-based algorithms predictable. Under a normal approximation (central limit theorem; CLT), the expected sample size to achieve a precision goal ω_{goal} is proportional to the per-observation variance $V(\theta)$ of the estimator (see Appendix C for the full derivation). As a concrete example, for a Bernoulli rate parameter θ as in the setting introduced in Section 2.4, $V(\theta) \propto \theta(1-\theta)$, giving

$$N_{\text{goal}}(\theta, \omega_{\text{goal}}) = \frac{4z_*^2}{\omega_{\text{goal}}^2}\,\theta(1-\theta) - 1, \tag{1}$$

where z_* is the critical value for the chosen credibility level (e.g., $z_* \approx 1.96$ for 95%). Because $\theta(1-\theta)$ is maximised at $\theta = 0.5$, the fair-coin scenario demands the largest sample for any given precision goal, and N_{goal} decreases monotonically as θ approaches 0 or 1. Figure 2 illustrates this dependence for the Bernoulli case across ω_{goal}.

This also illuminates DPitG's relationship to the HDI+ROPE algorithm: because the precision condition must be satisfied before any decision is possible, DPitG behaves approximately as the HDI+ROPE algorithm with $N_{\min} \approx N_{\text{goal}}$, replacing the ad-hoc floor (e.g. $N_{\min} = 30$) with a precision-derived bound.

For completeness, we mention that for continuous outcomes, $V(\theta)$ is replaced by the estimand variance σ^2; see Appendix C for the full derivation, including a Student-t variant for small samples.

2.4 Experimental Design

To evaluate and compare the performance of the three algorithms, we employ a synthetic data simulation based on Bernoulli trials (dichotomous data). While these methods are applicable to continuous data, we focus on the Bernoulli case to maintain comparability with the work of Kruschke (2015).[7]

We generate $M = 5,000$ independent sequences of $N_{\max} = 1,500$ trials each. This choice exceeds N_{goal} at the most demanding setting tested ($\omega_{\text{goal}} = 0.06$, $\theta_{\text{true}} = 0.5$), where Equation 1 gives $N_{\text{goal}} \approx 1{,}067$, ensuring that DPitG can reach its stopping criterion in the vast majority of experiments.

We test multiple scenarios for the ground truth probability $0.5 \leq \theta_{\text{true}} \leq 0.8$ against a null hypothesis of $\theta_{\text{null}} = 0.5$, including the fair-coin case $\theta_{\text{true}} = \theta_{\text{null}} = 0.5$.

Simulation Configuration

Unless stated otherwise, we use the following default parameters, chosen to reflect realistic experimental conditions (e.g., opinion polling):

(a) ROPE: Considering an effect size of ± 0.05 around the θ_{null}, we define $\text{ROPE}_{\text{max}}^{\text{min}} = [0.45, 0.55]$. ($\Delta_{\text{ROPE}} = 0.1$)

(b) Precision Goal $\omega_{\text{goal}} = 0.08$: I.e, we set $\omega_{\text{goal}} = 0.8 \cdot \Delta_{\text{ROPE}}$.

[7] We follow the setup of §13.3 of Kruschke (2015) and expand upon it with a broader range of θ_{true} scenarios, systematic performance metrics across $M = 5{,}000$ simulations, and visualisations designed to directly compare all three algorithms.

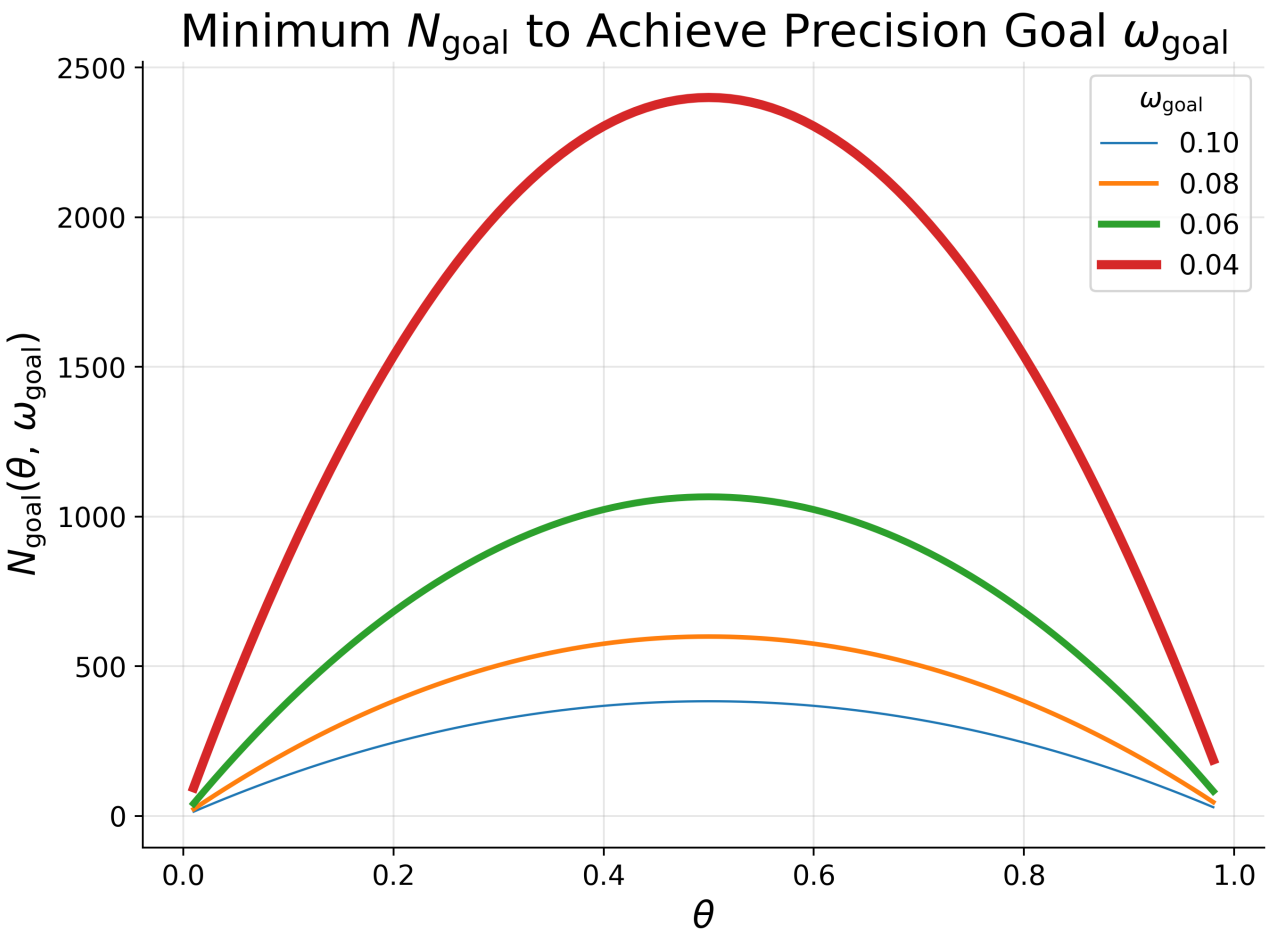


Figure 2: Expected minimum precision-based stop iteration $N_{goal}(\theta, \omega_{goal})$ as a function of θ for several values of ω_{goal} in the Bernoulli case. (Equation 1)

(c) **Minimum Sample Size ($N_{min} = 30$)**: A practical lower bound to guard against misleadingly decisive stops at small N, where aleatoric variability can drive the HDI into or out of the ROPE before sufficient data have accumulated (see Section 2.2.1). This parameter is required for the HDI+ROPE algorithm but not the precision-based algorithms.

(d) **Maximum Sample Size ($N_{max} = 1,500$)**: An upper bound for budgetary considerations. Experiments that are not decisive by this iteration are flagged as *indecisive*.

In the following section, we present the results of these simulations, first under these default settings and subsequently by varying key parameters to explore robustness.

3 Results

We present results in two parts. We open with the fair coin case ($\theta_{true} = \theta_{null} = 0.5$): a hand-picked experiment to illustrate the qualitative behaviour of all three algorithms (Section 3.1.1), followed by a large-scale study across $M = 5{,}000$ independent experiments (Section 3.1.2) and a sensitivity analysis over the precision goal ω_{goal} (Section 3.1.3). We then examine how all three algorithms perform as θ_{true} increases from 0.5 upwards, across and beyond the ROPE (Section 3.2).

3.1 Fair Coin ($\theta_{true} = 0.5$)

3.1.1 A Hand-Picked Sequence

Figure 3 presents a hand-picked randomly generated fair coin experiment chosen to illustrate contrasting outcomes across the three algorithms.[8] (note that all performance claims rest on the large-scale simulation study of

[8] This figure is similar to the top panels in Kruschke (2015) Figures 13.4 and 13.5.

Section 3.1.2) The posteriors in Figure 1 correspond to three specific iterations of this same experiment.[9]

Despite arising from the same sequence of coin tosses, the three algorithms reach sharply different decisions. The HDI+ROPE algorithm stops early at iteration 126 (vertical red dashed line) and, incorrectly, rejects θ_{null}. PitG stops at iteration 598 (first purple dot, where the precision goal is first met) but is left inconclusive, as the HDI straddles the ROPE boundary. DPitG waits until both conditions are jointly satisfied, stopping at iteration 804 (first green solid line coinciding with a purple dot) and correctly accepts θ_{null}. This divergence from a single experiment motivates the large-scale analysis that follows.

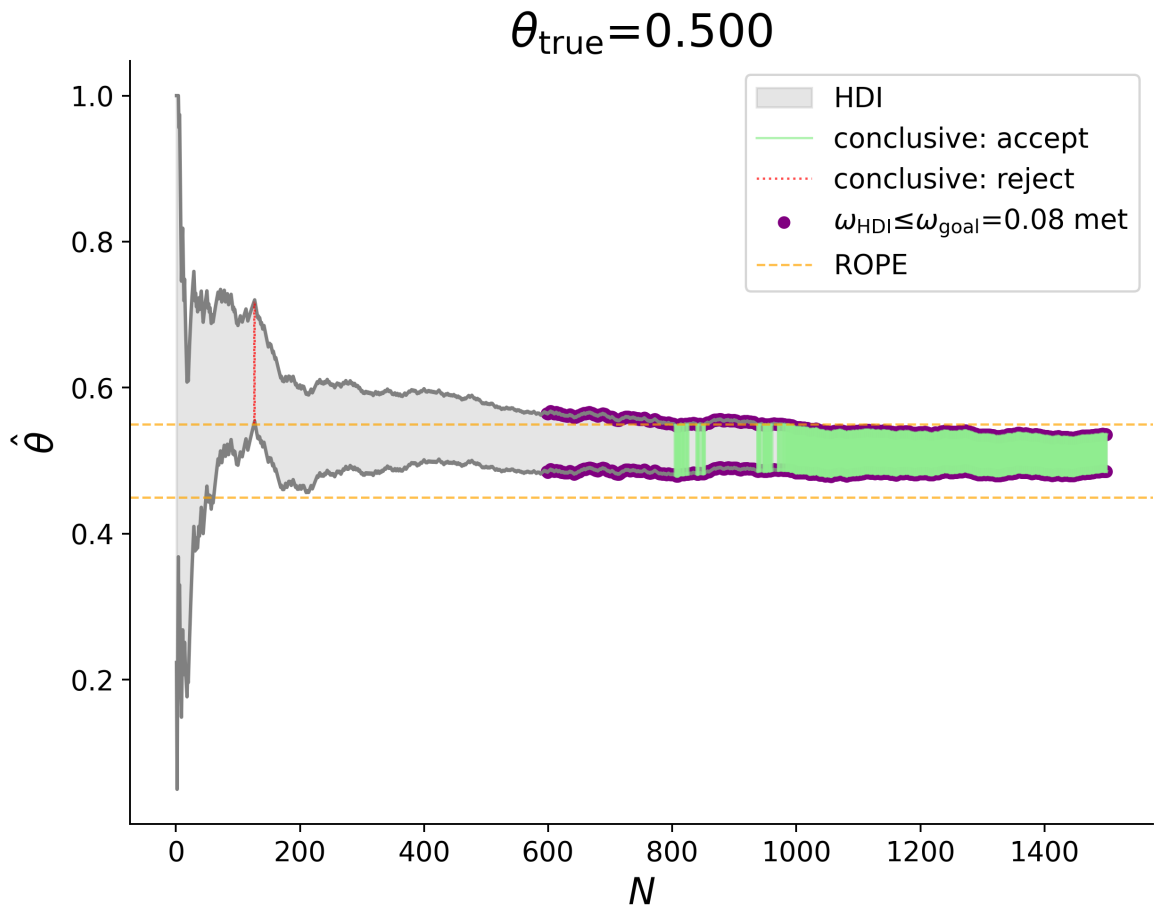


Figure 3: Cumulative sample proportion $\hat{\theta}(N)$ at each iteration N of a hand-picked randomly generated fair coin experiment ($\theta_{\text{true}} = \theta_{\text{null}} = 0.5$, $\omega_{\text{goal}} = 0.08$, $N_{\text{min}} = 30$) illustrating varied outcomes across sampling. $\hat{\theta}(N)$ denotes the observed proportion of successes (heads) in N trials. *Gray band*: 95% HDI of the posterior; its width is ω_{HDI}. *Horizontal dashed lines*: ROPE boundaries. *Vertical red dashed line*: Decision to reject θ_{null}. *Vertical green solid line*: Decision to accept θ_{null}. *Purple dots*: Precision goal met ($\omega_{\text{HDI}} \leq \omega_{\text{goal}}$). See Figure 1 for the posterior distributions at the three marked stop iterations.

3.1.2 Large-Scale Experiments

We expand this analysis to $M = 5{,}000$ experiments, displaying outcomes in Figures 4 and 5,[10] and summarising key statistics in Tables 2 and 3.

Figure 4 shows that the HDI+ROPE algorithm yields a wide range of stop iterations and sample proportions $\hat{\theta}$, with a non-negligible proportion of results falling outside the ROPE ($\hat{\theta}$ ranging from $\sim$0.25 to $\sim$0.80), leading to incorrect rejections of θ_{null}. Its N_{stop} distribution is skewed towards early stopping at ~ 400, well before $N_{\text{goal}} \sim 600$, with a long tail extending to later iterations.

PitG consistently stops near N_{goal}, with sample proportions clustered tightly around $\theta_{\text{true}} = 0.5$. However, many of these stops occur when the HDI straddles the ROPE boundary, yielding a high rate of inconclusive decisions (see Figure 5 and Table 2). DPitG exhibits a narrower $\hat{\theta}(N_{\text{stop}})$ distribution than PitG but extends to later iterations,

[9] The sequence may be found in the open-source material as `SEQUENCE_HANDPICKED`.

[10] Both of these visuals are inspired by Figure 13.6 of Kruschke (2015).

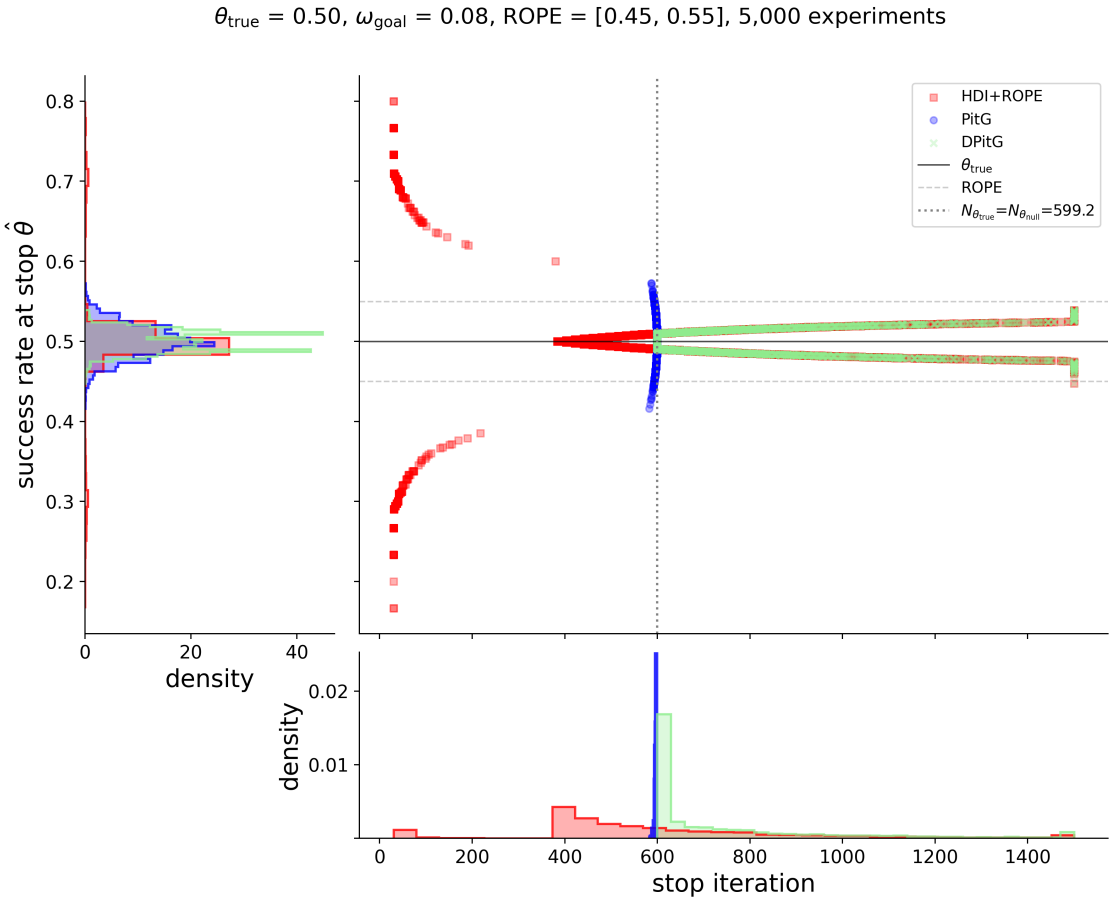


Figure 4: Outcomes of $M = 5{,}000$ fair coin experiments ($\theta_{\text{true}} = \theta_{\text{null}} = 0.5$, $\omega_{\text{goal}} = 0.08$, $N_{\text{min}} = 30$). *Main panel*: scatter plot of sample proportion $\hat{\theta}(N_{\text{stop}})$ vs. stop iteration N_{stop}. *Symbols*: red squares: HDI+ROPE; blue circles: PitG; green Xs: DPitG. *Solid horizontal line*: θ_{true}. *Horizontal dashed lines*: ROPE boundaries. *Vertical dashed line*: expected goal stop iteration N_{goal} (Equation 1); here $N_{\text{goal}}(\theta_{\text{true}}) = N_{\text{goal}}(\theta_{\text{null}})$ since $\theta_{\text{true}} = \theta_{\text{null}}$. *Bottom panel*: histograms of N_{stop}. *Left panel*: histograms of $\hat{\theta}(N_{\text{stop}})$. The marginal histograms reveal where most results concentrate, complementing the scatter plot which shows the correlations. Histogram binning varies by algorithm to best resolve each distribution.

achieving more conclusive results while maintaining zero estimation bias.[11] The N_{stop} histograms in Figure 4 make this concrete: DPitG's distribution begins near N_{goal}, precisely where PitG's terminates, confirming the structural equivalence noted in Section 2.3: DPitG behaves as the HDI+ROPE algorithm with $N_{\text{min}} \approx N_{\text{goal}}$ rather than a fixed small floor.

Figure 5 shows that the HDI+ROPE algorithm's rejection rate (left panel) quickly asymptotes to 6%, while its acceptance rate begins rising near $N \sim 400$, reflecting a gradual shift from incorrect rejections to correct acceptances as more data accumulate.

Both precision-based methods maintain a zero rejection rate (middle and right panels), as they require the precision criterion to be satisfied before any decision is made. For PitG (middle panel), the acceptance rate rises as a near step-function, jumping at N_{goal} to 37.9%, with the inconclusive rate settling at the complement 62.1%.

DPitG (right panel) presents a hybrid of the other two: it preserves the zero-rejection property of PitG while achieving a high conclusiveness rate akin to the HDI+ROPE algorithm, but with a more gradual transition from inconclusive to conclusive outcomes.

Table 2 summarises stop-iteration statistics across all M experiments; Table 3 then restricts to the conclusive subsets and breaks outcomes into acceptance and rejection rates alongside the $\hat{\theta}$ distribution at stop.

The HDI+ROPE algorithm achieves the highest conclusiveness rate (98.2%), but at the cost of a 6.3% false-reject rate. PitG delivers zero false rejections, but its conclusiveness rate of only 37.9% is strikingly low, with stop iterations all clustering at N_{goal}. DPitG delivers the balance sought in Section 2.2.3: a 97.8% conclusiveness rate

[11] In Section 3.2 we demonstrate that some bias emerges as symmetry breaks.

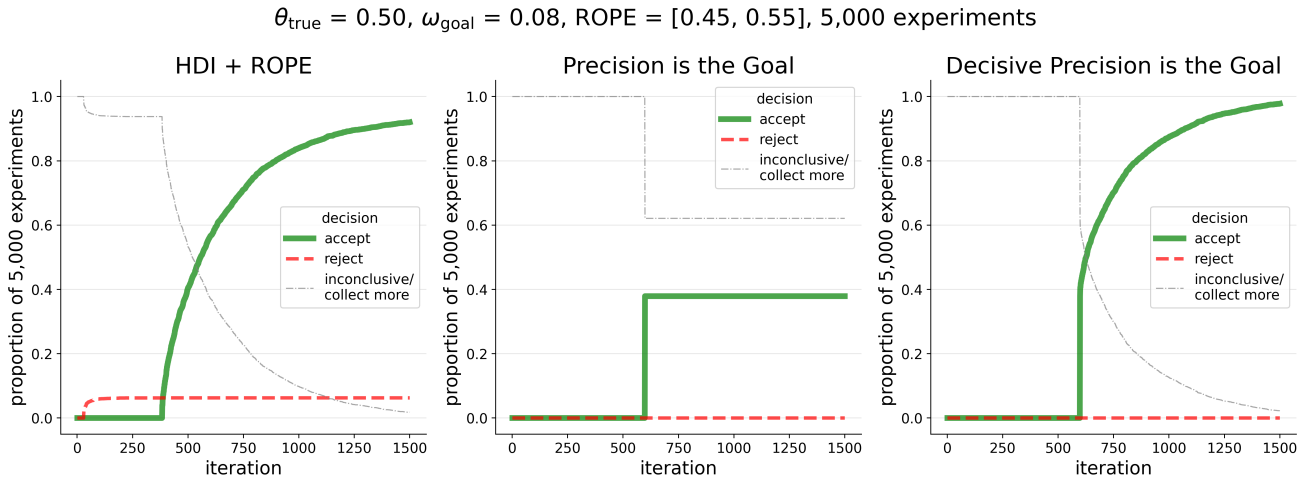


Figure 5: Cumulative decision rates across iterations for $M = 5{,}000$ fair coin experiments ($\theta_{\text{true}} = \theta_{\text{null}} = 0.5$, $\omega_{\text{goal}} = 0.08$, $N_{\min} = 30$). At each iteration N, the plotted proportions count all experiments that reached a decision at or before N; their sum is 100% at every point. *Left*: HDI+ROPE; *Middle*: Precision is the Goal; *Right*: Decisive Precision is the Goal. *Red dashed line*: reject θ_{null}. *Green solid line*: accept θ_{null}. *Gray dot-dashed line*: inconclusive.

Algorithm	Conclusive Rate	$N_{50\%}$	$N_{25\%}$–$N_{75\%}$	Δ_{IQR}
HDI+ROPE	0.982	521	415 - 723	308
PitG	**0.379**	599	598 - 599	1
DPitG	**0.978**	627	598 - 794	196

Table 2: Summary statistics for all $M = 5{,}000$ fair coin experiments ($\theta_{\text{true}} = \theta_{\text{null}} = 0.5$, $\omega_{\text{goal}} = 0.08$, $N_{\min} = 30$; cf. Figure 4). $N_{50\%}$: median stop iteration; $N_{25\%}$–$N_{75\%}$: interquartile range of stop iteration. Conclusive Rate values are color-coded: **red** flags a concerningly low rate (PitG); **green** highlights DPitG's strong rate with zero false rejections (cf. Table 3).

with zero false rejections. The lower quartile ($N_{25\%} = 598$) nearly equals $N_{\text{goal}} = 599$, confirming that a substantial fraction of DPitG experiments stop as soon as the precision goal is first met; the wider IQR of 598–794 reflects those that require additional iterations to satisfy the joint precision-and-conclusiveness criterion, at a median cost of only 4.7% more samples across all M experiments (4% when restricted to conclusive experiments; see Table 3) than PitG.

Algorithm	Conclusive	Acceptance Rate	Rejection Rate	$\hat{\theta}_{50\%}$	$\hat{\theta}_{25\%} - \hat{\theta}_{75\%}$
HDI+ROPE	4912	0.936	**0.064**	0.5000	0.4920-0.5081
PitG	**1895**	1	0	0.5008	0.4958-0.5041
DPitG	4888	1	0	0.5008	0.4895-0.5105

Table 3: Summary statistics for the conclusive subsets of experiments in Table 2. Conclusive: count of experiments (out of 5,000) with a conclusive outcome. Acceptance and Rejection rates are conditioned on conclusive experiments and therefore sum to 1. $\hat{\theta}_{50\%}$: median sample proportion at stop; $\hat{\theta}_{25\%} - \hat{\theta}_{75\%}$: interquartile range of sample proportion at stop. **Red** flags a non-zero false-reject rate (HDI+ROPE); **bold** flags a concerningly low conclusive count (PitG).

3.1.3 Sensitivity to ω_{goal}

The choice of ω_{goal} governs the fundamental precision–conclusiveness trade-off for PitG. Table 4 shows the effect across a range of values at $\theta_{\text{true}} = 0.5$. Tightening ω_{goal} (decreasing it) raises PitG's conclusiveness at the cost of a larger required sample size: N_{goal} grows quadratically as ω_{goal} decreases (Equation 1). DPitG's conclusiveness, by contrast, remains essentially flat across the full range (0.973–0.980), demonstrating robustness to the choice of ω_{goal}.

The conclusiveness ratio (DPitG/PitG) captures this contrast directly: it grows from 1.19 at $\omega_{\text{goal}} = 0.06$ to undefined at $\omega_{\text{goal}} = 0.10$, where PitG becomes entirely inconclusive. DPitG's robustness comes at the cost of

larger sample sizes, quantified by the $N_{50\%}/N_{\text{goal}}$ column: for $\omega_{\text{goal}} \leq 0.07$, the median DPitG stop equals N_{goal} (no premium); for looser goals the median sampling premium rises from 4% (cf. 4.7% across all M experiments) at $\omega_{\text{goal}} = 0.08$ to 41% at $\omega_{\text{goal}} = 0.10$.

At the tightest setting ($\omega_{\text{goal}} = 0.06$), the DPitG N_{stop} Δ_{IQR} is zero: most experiments stop at exactly N_{goal}, leaving no room for the additional iterations that DPitG normally uses to resolve borderline cases. At the loosest setting, when equal to the ROPE width ($\omega_{\text{goal}} = 0.10 = \Delta_{\text{ROPE}}$), PitG is always inconclusive, but DPitG still achieves 97.9% conclusiveness by collecting more data until both precision and conclusiveness are jointly satisfied.

The conclusiveness of DPitG is more sensitive to N_{max} (fixed at 1,500 in this example) than to ω_{goal}, whereas PitG always stops at N_{goal} regardless of N_{max}. Consequently, N_{goal} serves as a minimum budget for DPitG: for tight precision goals it is sufficient to achieve high conclusiveness; for looser goals, the researcher should set N_{max} generously to allow DPitG to reach its full potential.

		Conclusiveness			DPitG N_{stop}, conclusive experiments				
ω_{goal}	N_{goal}	PitG	DPitG	Ratio (DPitG/PitG)	$N_{25\%}$	$N_{75\%}$	Δ_{IQR}	$N_{50\%}$	$N_{50\%}/N_{\text{goal}}$
0.06	1,066	0.818	0.973	1.19	1,066	1,066	0	1,066	1.00
0.07	783	0.599	0.976	1.63	783	830	47	783	1.00
0.08	599	0.379	0.978	2.58	599	779	180	623	1.04
0.09	473	0.137	0.979	7.15	481	755	274	568	1.20
0.10	383	0.000	0.980	—	428	734	306	538	1.41

Table 4: Impact of ω_{goal} on PitG and DPitG performance ($\theta_{\text{true}} = \theta_{\text{null}} = 0.5$, $M = 5{,}000$ experiments, $N_{\text{max}} = 1{,}500$, $\Delta_{\text{ROPE}} = 0.1$). N_{goal}: expected precision-based stop iteration (Equation 1); Conclusiveness: fraction of M experiments with a conclusive outcome; Conclusiveness Ratio: DPitG/PitG conclusiveness (undefined at $\omega_{\text{goal}} = 0.10$ where PitG conclusiveness is zero); DPitG N_{stop}: quartile statistics restricted to conclusive DPitG experiments (results for $\omega_{\text{goal}} = 0.08$ differ slightly from Table 2, which summarises all experiments); $\Delta_{\text{IQR}} = N_{75\%} - N_{25\%}$. For $\omega_{\text{goal}} \leq 0.08$, $N_{25\%} = N_{\text{goal}}$, showing that at least a quarter of DPitG experiments stop as soon as the precision goal is first met; this is also reflected in the $N_{50\%}/N_{\text{goal}}$ column, which equals 1.00 for $\omega_{\text{goal}} \leq 0.07$.

The fair coin case thus demonstrates that DPitG combines the HDI+ROPE algorithm's high conclusiveness with PitG's zero-false-reject guarantee at a modest median cost. Section 3.2 shows that this advantage persists as θ_{true} departs from θ_{null}.

3.2 General Trends Across θ_{true} Values

Building on the fair coin baseline, we now examine how all three algorithms perform as θ_{true} increases from 0.5 upwards, across and beyond the ROPE, [12] with $\text{ROPE}_{\text{max}} = 0.55$ serving as a key inflection point. The primary case throughout is $\theta_{\text{null}} = 0.5$, where the null coincides with the precision maximum; we also include $\theta_{\text{null}} = 0.7$ in Figure 8 to examine behaviour when this symmetry breaks and the planned sample size is no longer a conservative upper bound. We present four complementary views via Figures 6–9: conclusiveness and decision rates, overall accuracy, estimation bias at stop, and a direct cost–benefit comparison of DPitG against PitG; qualitative trends generalise across ω_{goal} values (Section 3.1.3).

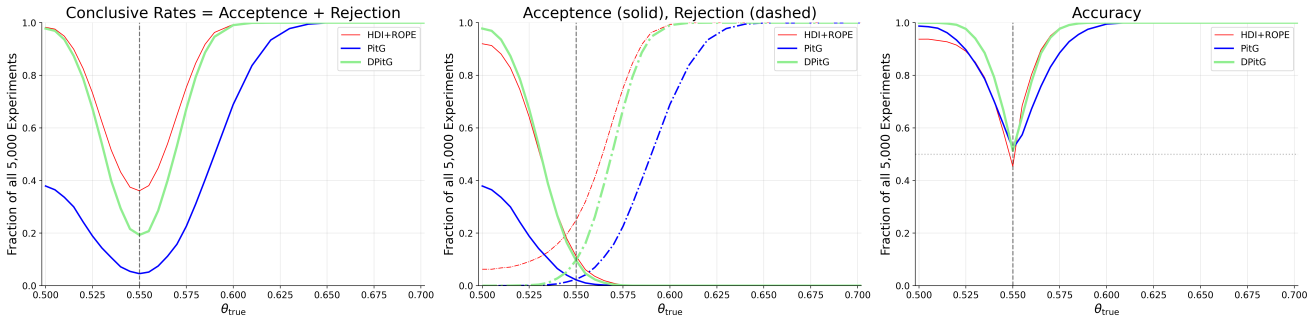


Figure 6: Algorithm comparison across θ_{true} values for $M = 5{,}000$ experiments ($\theta_{\text{null}} = 0.5$, $\omega_{\text{goal}} = 0.08$). *Colors*: red: HDI+ROPE; blue: PitG; green: DPitG. *Vertical dashed lines*: $\text{ROPE}_{\text{max}} = 0.55$. *Left*: conclusiveness rates. *Middle*: acceptance (solid lines) and rejection (dashed lines) rates; at each θ_{true} their sum equals the respective conclusiveness rate. *Right*: accuracy rates (proportion of correct decisions out of all M experiments); inconclusive experiments are assigned a decision based on whether $\hat{\theta}$ falls inside or outside the ROPE.

[12] By symmetry, results for $\theta_{\text{true}} < 0.5$ mirror those shown here, as reflected in Equation 1 and Figure 2.

The left panel of Figure 6 shows that all three algorithms show a dip in conclusiveness as θ_{true} approaches ROPE_{max}, after which conclusiveness rises again. The HDI+ROPE algorithm and DPitG recover to near 100% by $\theta_{\text{true}} = 0.6$; PitG only asymptotes to 100% at $\theta_{\text{true}} \approx 0.64$, at which point DPitG and PitG become effectively equivalent (for $\theta_{\text{null}} = 0.5$).

The precision-based methods are deliberately conservative near the ROPE boundaries, yielding lower conclusiveness rates than the HDI+ROPE algorithm in this region; at ROPE_{max} itself, DPitG is conclusive in only 19% of experiments and PitG in only 5%. This dip is a natural consequence of the Decision Rule (see Section 2.1.4): the ROPE boundaries are exactly the region where neither an accept nor a reject verdict is easily justified.

Breaking the conclusiveness into its acceptance and rejection components (middle panel of Figure 6) reveals the error structure. For $\theta_{\text{null}} = 0.5$, acceptance is the correct decision within the ROPE ($\theta_{\text{true}} \leq 0.55$) and rejection is correct outside it. The HDI+ROPE algorithm's false-reject rate climbs from $\sim$6% at $\theta_{\text{true}} = 0.5$ to $\sim$30% at ROPE_{max}; its false-accept rate peaks at $\sim$10% at the ROPE boundary before dropping to 0% by $\theta_{\text{true}} = 0.57$. Both precision-based methods maintain a near-zero false-reject rate throughout: exactly zero well within the ROPE interior, rising to a small level near the boundary: DPitG reaching $\sim$10%, and PitG lower only because its conclusiveness rate is much smaller there. Both also have false-accept rates that mirror the false-reject rates around the ROPE boundary, which drops quickly as θ_{true} moves outside.

The right panel of Figure 6 shows overall *accuracy*, the proportion of correct decisions across all M experiments. Since the three algorithms have different conclusiveness rates, inconclusive experiments must be assigned a decision for comparison. We adopt the natural default: accept if $\hat{\theta}$ lies within the ROPE, reject otherwise, the rule a practitioner would apply informally using the point estimate alone, and the most common approach in empirical settings where credible intervals are not examined; a more critical treatment of alternative imputation rules is deferred to future work.

All three algorithms reach their minimum accuracy at ROPE_{max}. Within the ROPE, DPitG consistently leads PitG, reaching 99.5% at $\theta_{\text{true}} = 0.5$ compared to PitG's 98.5%. Outside the ROPE, the HDI+ROPE algorithm and DPitG reach 100% by $\theta_{\text{true}} = 0.58$, whereas PitG achieves this only at $\theta_{\text{true}} \approx 0.60$.

That the HDI+ROPE algorithm reaches 100% accuracy at $\theta_{\text{true}} = 0.58$ despite producing biased $\hat{\theta}$ estimates motivates a closer look at the observed sample proportion statistics. Figure 7 shows, for conclusive experiments, the IQR of $\hat{\theta}(N_{\text{stop}})$ (left panel) and the median bias $\hat{\theta}_{50\%} - \theta_{\text{true}}$ (right panel).[13]

As discussed in Section 2.1.4, accept/reject decisions rest on the HDI's position relative to the ROPE, not on $\hat{\theta}$; a bias in $\hat{\theta}$ does not invalidate the decision so long as the HDI falls cleanly within (or outside) the ROPE.

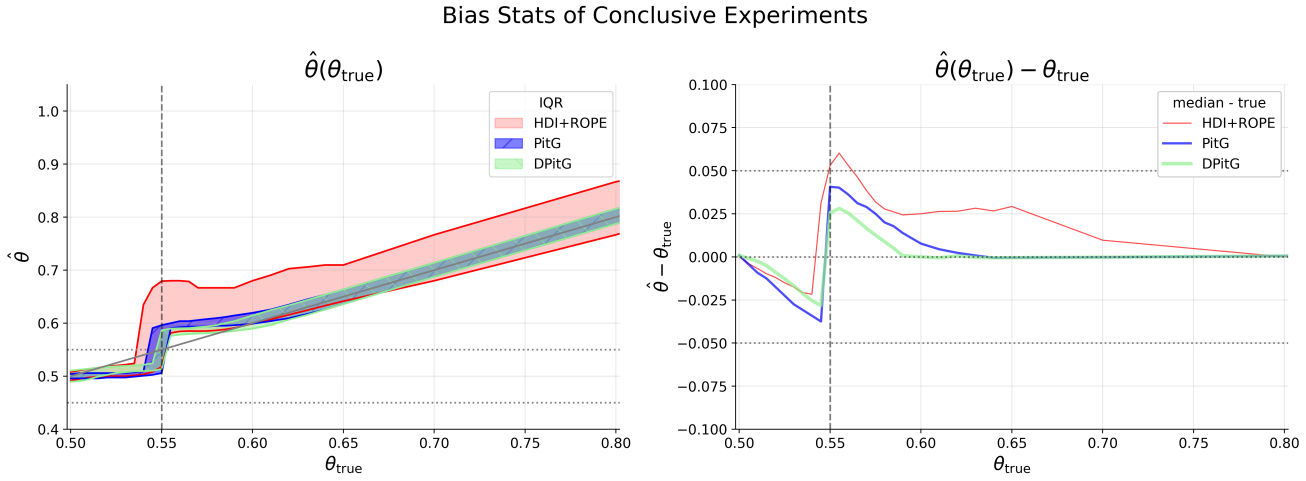


Figure 7: Sample proportion $\hat{\theta}(N_{\text{stop}})$ at the stop iteration as a function of θ_{true}, for conclusive experiments only (subsets of $M = 5{,}000$, $\theta_{\text{null}} = 0.5$, $\omega_{\text{goal}} = 0.08$, $N_{\text{min}} = 30$). *Colors*: red: HDI+ROPE; blue: PitG; green: DPitG. *Horizontal dotted lines*: ROPE boundaries. *Vertical dashed line*: ROPE_{max}. *Left*: interquartile range $\hat{\theta}_{25\%}$–$\hat{\theta}_{75\%}$ of $\hat{\theta}(N_{\text{stop}})$. *Diagonal line*: parity line $\hat{\theta} = \theta_{\text{true}}$. *Right*: median bias $\hat{\theta}_{50\%} - \theta_{\text{true}}$.

The HDI+ROPE algorithm's $\hat{\theta}$ bias is therefore more telling as a symptom of its lack of a precision requirement than as a decision-quality metric. At $\theta_{\text{true}} = 0.5$, all three algorithms show zero bias, as expected from the symmetry of the fair coin case. Once θ_{true} deviates from 0.5, the HDI+ROPE algorithm's bias grows markedly: its median bias approaches ROPE_{max}, with the upper IQR extending substantially further, because the HDI+ROPE algorithm can stop on skewed samples before sufficient data have accumulated to constrain the posterior.

The precision-based methods exhibit substantially smaller $\hat{\theta}$ bias, well within the ROPE width across all θ_{true} values, as a natural consequence of their precision requirement. Among conclusive experiments within the ROPE,

[13] We find similar trends for the mean, but omit it to reduce clutter.

PitG shows higher bias than DPitG; for inconclusive experiments (not shown[14]) (where PitG contributes far more cases than DPitG) the pattern reverses, PitG's bias being slightly lower. Pooled across both subsets, the aggregate difference is negligible: both methods yield near-zero bias at all θ_{true} values, with DPitG's aggregate bias only marginally higher than PitG's. Despite this, DPitG decision accuracy exceeds PitG's at every θ_{true} value (cf. Figure 6, right panel), because its substantially higher conclusiveness rate more than compensates.

DPitG is decisively superior to PitG near and within the ROPE, at the cost of a larger median N_{stop}; having characterised the estimation properties of that superiority, we now quantify its sampling cost. Figure 8 quantifies N_{stop} for all three algorithms, restricting to conclusive outcomes since inconclusive experiments always stop at $N_{\max}$ and their stop iteration therefore carries no information about algorithm behaviour; including the HDI+ROPE algorithm also illustrates why the precision requirement matters.

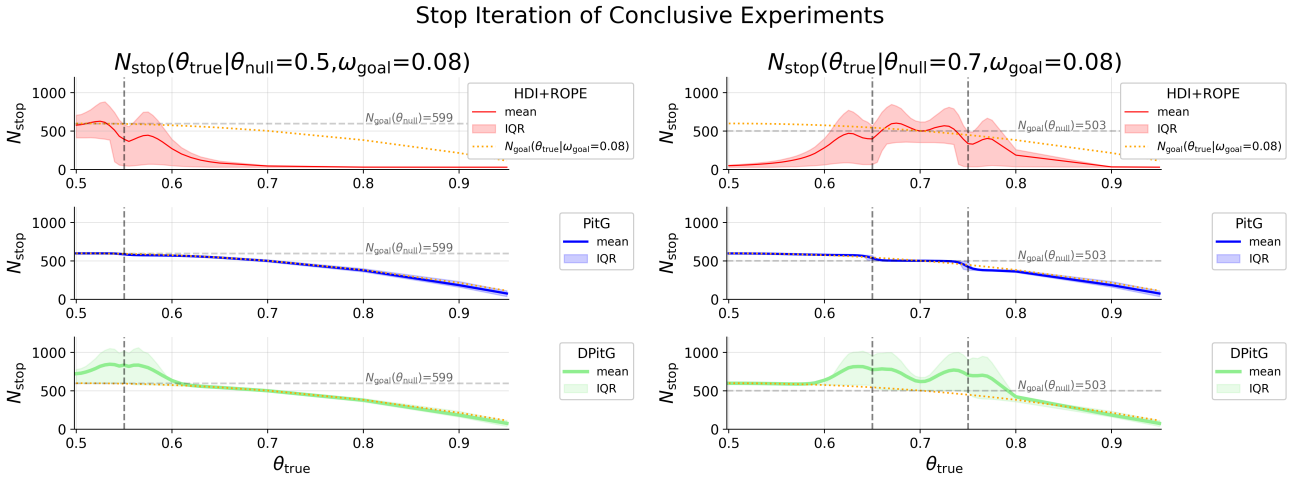


Figure 8: Stop iteration N_{stop} statistics for conclusive experiments across θ_{true} values ($\omega_{\text{goal}} = 0.08$). *Left*: $\theta_{\text{null}} = 0.5$; *Right*: $\theta_{\text{null}} = 0.7$. In each column the top, middle, and bottom panels correspond to HDI+ROPE, PitG, and DPitG, respectively. *Band*: interquartile range $N_{25\%}$–$N_{75\%}$ of N_{stop}. *Solid line*: median $N_{50\%}$. *Dotted line*: true precision-goal iteration $N_{\text{goal}}(\theta_{\text{true}}, \omega_{\text{goal}})$ (Equation 1). *Horizontal dashed line*: null precision-goal iteration $N_{\text{goal}}(\theta_{\text{null}}, \omega_{\text{goal}})$, the researcher's planned sample size.

The HDI+ROPE algorithm shows distinctly different behaviour inside and outside the ROPE. Within the ROPE, N_{stop} loosely tracks $N_{\text{goal}}(\theta_{\text{true}})$, but with a wide IQR reflecting the absence of a precision criterion. Right at $\text{ROPE}_{\max}$, a distinctive V-shaped signature is visible: N_{stop} dips to a narrow trough at the boundary, as the few conclusive experiments there stop early by catching a chance excursion of the HDI just outside the ROPE. Outside the ROPE, N_{stop} falls well below $N_{\text{goal}}(\theta_{\text{true}})$: The HDI+ROPE algorithm correctly rejects θ_{null} much faster than precision-based methods, but does so on the basis of imprecise posteriors, producing the strong $\hat{\theta}$ bias (cf. Figure 7).

PitG stops tightly around $N_{\text{goal}}(\theta_{\text{true}})$ by design, making it the natural reference for understanding the precision cost. The contrast with the planned quantity $N_{\text{goal}}(\theta_{\text{null}})$ reveals an important asymmetry: because $N_{\text{goal}}(\theta)$ is maximised at $\theta = 0.5$ and decreases as θ moves towards 0 or 1, the direction of the gap depends on where θ_{true} lies relative to θ_{null}.

For $\theta_{\text{null}} = 0.5$, any deviation of θ_{true} from 0.5 implies $N_{\text{goal}}(\theta_{\text{true}}) < N_{\text{goal}}(\theta_{\text{null}})$: the expected stop iteration of $N_{\text{goal}} = 599$ is a conservative upper bound on N_{stop}, and experiments stop earlier whenever the truth differs from the null.

The right column of Figure 8 shows what happens when the null does not coincide with the precision maximum. The case $\theta_{\text{null}} = 0.7$ is representative of a researcher expecting an elevated base rate (such as a presumed 70% success probability for an established treatment) who must judge whether the true rate is practically equivalent to that benchmark. Unlike the fair coin case, where $\theta_{\text{null}} = 0.5$ maximises $\theta(1-\theta)$ so any deviation of θ_{true} reduces the required sample, here the symmetry breaks: the same deviation can either reduce or increase N_{goal} depending on its direction.

In particular, when $\theta_{\text{true}} < \theta_{\text{null}}$ (moving closer to 0.5), $N_{\text{goal}}(\theta_{\text{true}}) > N_{\text{goal}}(\theta_{\text{null}} = 0.7) \approx 503$: the planned budget underestimates the samples required, and $N_{\text{goal}}(\theta_{\text{null}})$ acts as a lower bound rather than an upper one.[15]

[14] Bias statistics for inconclusive experiments and the pooled analysis are available in the open-source materials.

[15] When $\theta_{\text{true}} > \theta_{\text{null}} = 0.7$ (moving further from 0.5), $N_{\text{goal}}(\theta_{\text{true}}) < N_{\text{goal}}(\theta_{\text{null}})$ and experiments stop sooner than planned. The PitG median stop iteration visibly exceeds $N_{\text{goal}}(\theta_{\text{null}}) = 503$ for $\theta_{\text{true}} < 0.7$, confirming this underestimate. DPitG shows qualitatively similar patterns to the $\theta_{\text{null}} = 0.5$ column near both ROPE boundaries, elevated above $N_{\text{goal}}(\theta_{\text{true}})$ where the conclusiveness gain is largest.

In general, $N_{\text{goal}}(\theta_{\text{null}})$ serves as the researcher's advance planning estimate: a conservative upper bound when θ_{null} lies at the precision maximum (e.g. $\theta_{\text{null}} = 0.5$), and a floor otherwise. The researcher should set $N_{\max}$ according to their resources, with $N_{\text{goal}}(\theta_{\text{null}})$ as the baseline (here $N_{\max} = 1{,}500$).

DPitG's N_{stop} equals or exceeds PitG's within the ROPE and in the vicinity of its boundaries, the extra iterations being precisely what accounts for the higher conclusiveness seen in Figure 6. Once θ_{true} is far enough from the ROPE, DPitG converges to PitG and both stop at $N_{\text{goal}}(\theta_{\text{true}})$.

Figure 9 places the DPitG–PitG trade-off in direct comparison by overlaying the ratio of conclusiveness rates (from Figure 6) and the ratio of N_{stop} (from Figure 8) across θ_{true}.[16]

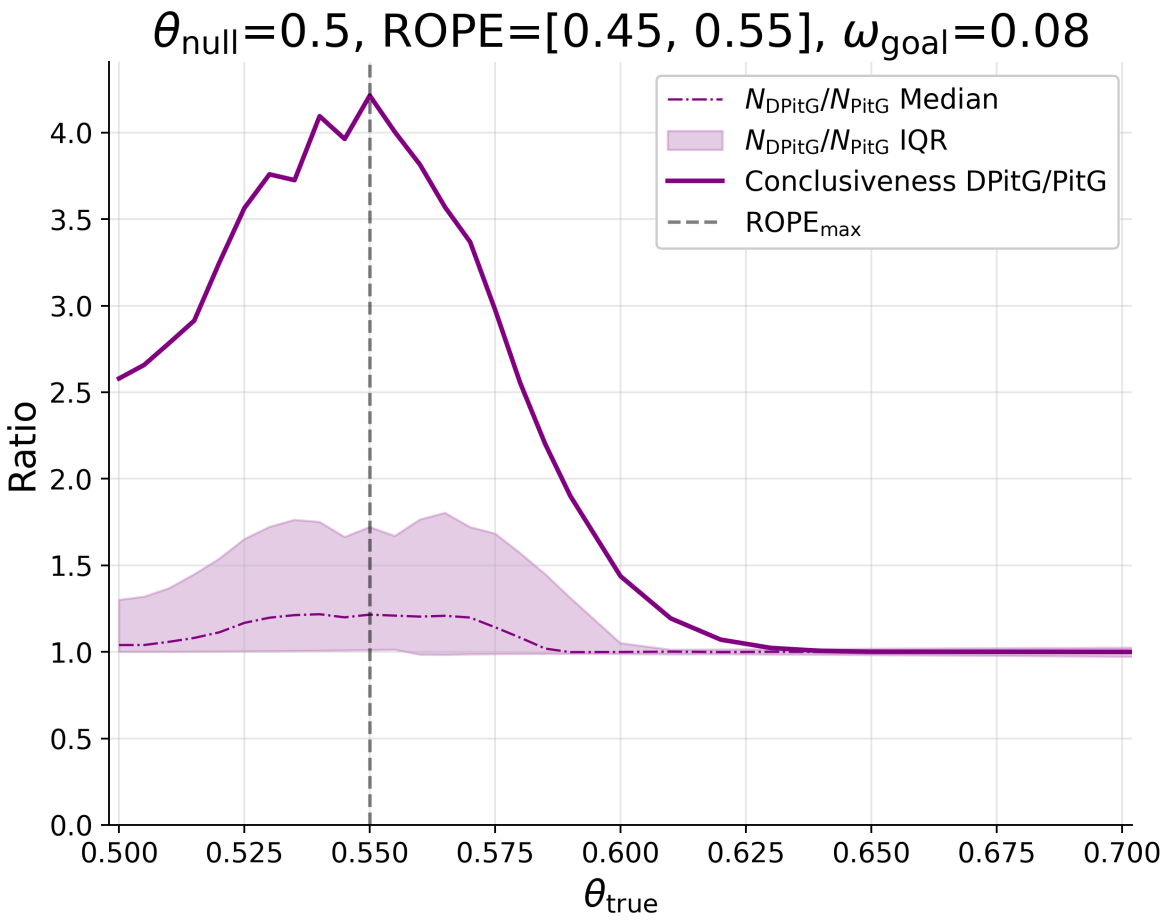


Figure 9: DPitG-to-PitG ratios of conclusiveness rate and stop iteration across θ_{true} values ($M = 5{,}000$, $\theta_{\text{null}} = 0.5$, $\omega_{\text{goal}} = 0.08$). *Solid line*: conclusiveness ratio (cf. Figure 6). *Dot-dashed line*: median N_{stop} ratio. *Band*: interquartile range of the N_{stop} ratio. *Vertical dashed line*: $\text{ROPE}_{\max} = 0.55$.

The conclusiveness ratio substantially exceeds the N_{stop} ratio throughout, particularly within and near the ROPE. Strikingly, the conclusiveness ratio remains above 1 even where the N_{stop} ratio approaches 1 (roughly $0.60 \lesssim \theta_{\text{true}} \lesssim 0.63$), confirming that DPitG gains conclusive outcomes at negligible extra sampling cost in this region. The lower IQR band confirms that a substantial fraction of DPitG experiments stop exactly when PitG does. The fair coin quantitative summary (a 2.6× conclusiveness gain at a median 4% sampling premium) is consistent with Table 2 and Section 3.1.2; Table 4 shows how the conclusiveness ratio varies with ω_{goal}.

The DPitG advantage documented here generalises: qualitatively similar conclusiveness gains hold for $\theta_{\text{null}} = 0.7$ (as shown in the right panel of Figure 8) and across ω_{goal} values (see Section 3.1.3 for the fair coin case).

4 Discussion

DPitG resolves the tension inherent in PitG, requiring both precision and decisiveness simultaneously, at a modest median sampling cost (Sections 3.1.2–3.2). We now place this result in methodological context, draw out its

[16] The apparent flattening of the stop iteration ratio near $\text{ROPE}_{\max}$ is an artefact of the finite cap $N_{\max} = 1{,}500$; with a larger cap both curves would peak sharply at the boundary.

practical implications, and candidly acknowledge its limitations.

DPitG operates under a different philosophy from classical sequential methods. Group sequential designs (Jennison and Turnbull, 2000) control the familywise Type I error rate across interim analyses via alpha-spending functions, and are therefore tied to the number and timing of planned looks. DPitG requires no such correction, since the Bayesian posterior accounts for all accumulated data without reference to the stopping intention. This property extends to simultaneous tests: the DPitG stopping rule applies unchanged across independent parallel experiments without multiple-comparisons correction; Appendix E develops this in full, together with all three types of Bayesian power analysis (prospective, retrospective, and replication).

Closest in spirit among Bayesian sequential designs is Bayes Factor Design Analysis (BFDA; Schönbrodt and Wagenmakers 2018), which plans sample size to achieve a target Bayes Factor with specified probability.[17] The key distinction is that BFDA targets evidential strength (ratio of marginal likelihoods) whereas DPitG targets posterior precision; the two are complementary as planning tools, and a design requiring both a conclusive HDI+ROPE framework verdict and a minimum Bayes Factor is a natural extension.

While comparisons with competing sequential methods clarify DPitG's positioning, its computational scope deserves equal attention. DPitG is a generic sequential stopping rule applicable to any posterior admitting a credible interval, including non-conjugate and hierarchical settings that require MCMC; the present analysis restricts to conjugate posteriors, where the HDI is fully analytical and no specialist probabilistic programming software is required.

A recurring concern about PitG (and by extension DPitG) is that the precision requirement can seem too demanding in practice. Researchers frequently underestimate the noise in their measurements and are consequently surprised by the sample sizes that ω_{goal} implies (J.K. Kruschke, pers. comm. 2022). This is not a flaw in the method but a form of epistemic honesty: imprecise posteriors produce inconclusive results regardless of the algorithm applied; DPitG simply refuses to declare a verdict until the posterior is genuinely informative. Setting ω_{goal} thus forces realistic planning rather than masking insufficient power behind a false reject (i.e., false positive) or a silently inconclusive stop. This discipline is a feature, not a limitation, for any analyst whose conclusions must rest on genuinely informative data.

DPitG serves researchers in two complementary roles. As an operational stopping rule, it is best suited to settings where data collection is relatively cheap and the budget comfortably exceeds N_{goal}: large-scale digital A/B testing (Johari et al., 2022), opinion polling, and automated quality-control pipelines are typical examples. As a planning tool, it provides a diagnostic function even in resource-constrained settings such as longitudinal studies, clinical trials, or expensive policy evaluations: N_{goal} makes explicit what precision is achievable within the available budget, enabling an honest assessment of what conclusions the data can and cannot support, both prospectively (before collection begins) and retrospectively (to audit the precision of a completed study).

The ROPE must be grounded in domain knowledge rather than chosen to suit the data (Kruschke, 2018); Appendix G discusses principled approaches for setting it, including the MCID, bioequivalence standards, and sensitivity analyses, together with three broader considerations supporting DPitG's adoption (equivalence testing, pre-registration, and the replication crisis). Together with ω_{goal} and $N_{\max}$, the ROPE completes the three design parameters that must be fixed before data collection begins.

With those design inputs committed, the joint operating characteristics of DPitG can be assessed prospectively: a researcher can check, analytically or via simulation, that the planned $N_{\max}$ achieves the target conclusiveness rate at the effect sizes of interest. As the Results demonstrate, error rates are negligible except near the ROPE boundaries, where neither an accept nor a reject verdict is easily justified. A key practical implication (Section 3.1.3) is that conclusiveness is more sensitive to $N_{\max}$ than to ω_{goal}: for the fair coin case ($\Delta_{\text{ROPE}} = 0.1$), the median sampling premium relative to N_{goal} is zero for $\omega_{\text{goal}} \leq 0.07$ and rises to 41% at $\omega_{\text{goal}} = 0.10$, so setting $N_{\max}$ generously is the primary practical lever for ensuring high conclusiveness across a range of precision goals.

We now turn to the limits of the present work. Four limitations deserve acknowledgement. First, the results are sensitive to the three design parameters (the ROPE, ω_{goal}, and $N_{\max}$), all of which must be committed to before data collection begins. A narrower ROPE (and with it a tighter ω_{goal}) increases the sample size required for decisiveness and raises the inconclusive rate near its boundaries. Second, the stopping properties are validated empirically only for binary (Bernoulli) data under a conjugate Beta prior. The normal approximation underpinning N_{goal} (Equation 1) may underestimate the required sample size in small-sample settings or for highly skewed likelihoods; in such cases the exact Binomial result in Appendix C should be preferred. Third, the present analysis uses a flat (uniform) prior throughout, as a conservative choice that avoids imposing unverified assumptions. The flat prior contributes no pseudo-observations, requiring more data to achieve ω_{goal} than a well-justified informative prior would (Gelman et al., 2013, Ch. 2). Informative priors, where available and well-justified, can therefore serve as a practical mitigation by narrowing the starting HDI width and lowering the required sample size. Conversely, a materially incorrect informative prior may bias the posterior in a way that the precision criterion alone cannot detect (Gelman et al., 2013, Ch. 6); we defer a full treatment of principled prior selection to a later study. Fourth, while DPitG applies in principle to any posterior admitting such an interval, this paper does not characterise its

[17] This is a prospective planning use of the BF, distinct from using the BF as a sequential stopping rule (see Appendix F for a discussion of BF stopping and its limitations).

behaviour when the HDI is estimated from MCMC samples. In such settings the HDI estimate carries Monte Carlo error (Kruschke, 2015) that could affect stopping calibration, particularly in early data collection when the posterior is diffuse, and this remains an open question for subsequent investigation.

5 Conclusion

We propose DPitG as a sequential stopping rule that jointly requires posterior precision and a conclusive verdict before data collection ends. PitG eliminates the early-peeking bias of coupled methods by decoupling the stopping and decision criteria (Kruschke, 2015), but accepts high inconclusive rates as the price. Inconclusive rates rise steeply the closer the null hypothesis is to the true value.

DPitG closes this gap by imposing both requirements simultaneously: data collection continues until $\omega_{\text{HDI}} \leq \omega_{\text{goal}}$ *and* the posterior's position relative to the ROPE yields a conclusive verdict. This joint requirement delivers a 2.6× gain in conclusiveness over PitG at a median cost of only 4.7% (4% among conclusive experiments) more samples ($\theta_{\text{true}} = 0.5$, $\omega_{\text{goal}} = 0.08$, $\text{ROPE}_{\text{min}}^{\text{max}} = [0.45, 0.55]$), while the HDI+ROPE algorithm accrues a systematic false-positive rate because its coupled design can stop on wide, early posteriors before they are representative. The gain is robust: DPitG's conclusiveness holds nearly flat (97.3%–98.0%) across the same ω_{goal} range, over which PitG's conclusiveness varies widely[18] and is largest near the ROPE boundaries where evidence is hardest to read.

Unlike NHST, which can only reject or fail to reject the null (see Appendix B), the HDI+ROPE framework's decision rule enables positive *acceptance* of the null when the entire HDI falls within the ROPE, a critical advantage for confirmatory research areas such as bioequivalence, safety testing, and software regression testing; DPitG reliably delivers this verdict.

Prospective planning is straightforward: the closed-form $N_{\text{goal}}(\theta_{\text{null}}, \omega_{\text{goal}}) \propto \theta_{\text{null}}(1 - \theta_{\text{null}})/\omega_{\text{goal}}^2$ provides the minimum budget required, with N_{max} set generously above this baseline to ensure DPitG reaches its full conclusiveness potential.

The framework extends to single-group continuous data (Appendix C) and to two-group comparisons, both binomial and continuous (Appendix D).[19] All algorithms are implemented in openly available code[20], with an interactive online calculator for prospective sample-size planning and retrospective analysis available at https://r-w-t-y.streamlit.app.

DPitG is the method of choice whenever a reliable verdict (acceptance, rejection, or inconclusive) is required and the budget is sufficient to reach $N_{\text{goal}}(\theta_{\text{null}})$; the closed-form N_{goal} makes this judgement transparent before any data are collected.

A HDI and ETI: Credible Interval Variants

Given a posterior density $p(\theta \mid \mathbf{x})$, where θ is the parameter of interest and $\mathbf{x}$ the observed data, a $(1-\alpha)$ credible interval $[L, U]$ satisfies

$$\int_L^U p(\theta \mid \mathbf{x})\, d\theta = 1 - \alpha. \tag{2}$$

Infinitely many intervals satisfy this constraint. This paper uses the Highest Density Interval throughout; the Equal-Tailed Interval is described below for comparison.

Highest Density Interval (HDI)

The HDI is the shortest interval satisfying Equation 2:

$$[L^*, U^*] = \underset{[L,U]:\ \int_L^U p(\theta|\mathbf{x})\, d\theta = 1-\alpha}{\arg\min} (U - L). \tag{3}$$

An equivalent characterisation for unimodal posteriors: $[L^*, U^*]$ is the unique interval for which $p(L^* \mid \mathbf{x}) = p(U^* \mid \mathbf{x})$, so every point inside has posterior density at least as high as every point outside (hence "highest density").[21] For the Binomial model used in this work, under a flat prior Beta(1, 1), the posterior after s successes and f failures is Beta($s+1$, $f+1$), and the HDI is found by minimising the interval width over the lower tail probability.[22]

[18] And collapses to zero when $\omega_{\text{goal}} = \text{ROPE}_{\text{max}} - \text{ROPE}_{\text{min}}$.

[19] These extensions are theoretical derivations; stopping-property validation beyond the single-group Bernoulli case remains to be completed in a dedicated simulation study.

[20] https://github.com/elzurdo/dpitg

[21] For multimodal posteriors, multiple intervals can satisfy the equal-density boundary condition, and the HDI may be a union of disjoint intervals; the shortest-interval definition in Equation 3 selects the one with minimum total width. All posteriors in this paper are unimodal.

[22] Specifically, minimise $F^{-1}(1 - \alpha + \ell) - F^{-1}(\ell)$ over $\ell \geq 0$; the minimiser ℓ^* gives $L^* = F^{-1}(\ell^*)$ and $U^* = F^{-1}(1 - \alpha + \ell^*)$ (`HDIofICDF` in the accompanying code).

Equal-Tailed Interval (ETI)

The ETI places $\alpha/2$ probability mass in each tail:

$$L_{\text{ETI}} = F^{-1}\left(\tfrac{\alpha}{2}\right), \qquad U_{\text{ETI}} = F^{-1}\left(1 - \tfrac{\alpha}{2}\right), \tag{4}$$

where F^{-1} is the posterior quantile function. The ETI is simple to compute and uniquely defined, but it is not generally the shortest interval. For symmetric, unimodal posteriors the HDI and ETI coincide; they diverge for skewed distributions (for instance, a Beta posterior near $\theta = 0$ or $\theta = 1$) where the HDI is noticeably shorter and therefore preferred.

B Regarding NHST

Null hypothesis significance testing (NHST, Fisher 1925; Neyman and Pearson 1933) is included here as contextual background rather than as a competing method. A p-value measures the probability of observing data at least as extreme as those obtained *assuming the null is true*, $P(\text{data} \mid H_0)$, not the probability that the null is true given the data, $P(H_0 \mid \text{data})$ (Cohen, 1994; Wasserstein and Lazar, 2016); consequently, NHST can only reject the null or remain inconclusive; it carries no mechanism to positively accept it, making it structurally unsuitable for the precision-and-decisiveness criterion that motivates DPitG. Understanding its sequential behaviour nevertheless clarifies why the Bayesian HDI+ROPE framework was developed.

In NHST, at each iteration the experiment stops and θ_{null} is rejected whenever the p-value falls below the false positive rate threshold α_{FPR}. Since the decision rule is based solely on this single threshold, data that fail to trigger rejection yield an inconclusive result rather than evidence in favour of the null.

Using the same fair coin setup ($\theta_{\text{true}} = \theta_{\text{null}} = 0.5$, $N_{\min} = 30$) described in the Methods section (2), we apply NHST with $\alpha_{\text{FPR}} = 0.05$ as the coupled stopping and decision criterion. A reliable sequential method should maintain a low rejection rate throughout when the null hypothesis is true: as demonstrated in Section 3.1.2, the HDI+ROPE algorithm's rejection rate quickly asymptotes near 6% and both precision-based methods maintain a zero rejection rate.

Figure 10 shows the opposite for NHST: the rejection rate (red solid) grows steadily over time, while the inconclusive proportion (gray dashed) decreases. To make this trend visible, the simulation is extended to $N_{\max} =$ 30,000 iterations, well beyond the $N_{\max} = 1{,}500$ used in the main analysis, since false rejections accumulate slowly and the growing trend would not be apparent at shorter horizons.

As pointed out by Kruschke (2015), as evidence accumulates under NHST the probability of observing a sample extreme enough to trigger rejection grows without bound, even when the null hypothesis is true. To paraphrase: "with infinite peeking, the false positive rate approaches 100%." This is because the stopping rule is tied directly to the decision rule with no reference to an effect size: the more data collected, the more likely an extreme sample will appear. Continuously monitoring the data and applying a fixed p-value threshold therefore leads to an inflated false-positive (Type I) error rate, a manifestation of the multiple-testing problem (Simmons et al., 2011). Lowering α_{FPR} delays but does not prevent this accumulation: for any fixed threshold, the false rejection rate still grows without bound as data continue to be collected.

C Derivation of Expected Precision Goal Stop Iteration

The precision-based stopping algorithms (PitG Kruschke 2015 and DPitG) halt when the HDI width (see Appendix A) falls below the precision goal ω_{goal}. Because the posterior HDI width shrinks as $N^{-1/2}$, one can predict in advance the iteration at which this threshold will be crossed. The General CLT Form below uses a normal approximation and is intended as an analytical guide; the Binomial Case refines this using the exact Beta posterior variance, and the Continuous Case provides two variants (CLT and Student-t).

General CLT Form

Assume an estimator $\hat{\theta}_N$ with asymptotic distribution

$$\hat{\theta}_N \approx \mathcal{N}\left(\theta, \frac{V(\theta)}{N}\right), \tag{5}$$

where $V(\theta)$ is the per-observation variance contribution. For a symmetric central interval with critical value z_* (e.g., $z_* = 1.96$ for 95%), the interval width is approximately

$$w \approx 2z_*\sqrt{\frac{V(\theta)}{N}}. \tag{6}$$

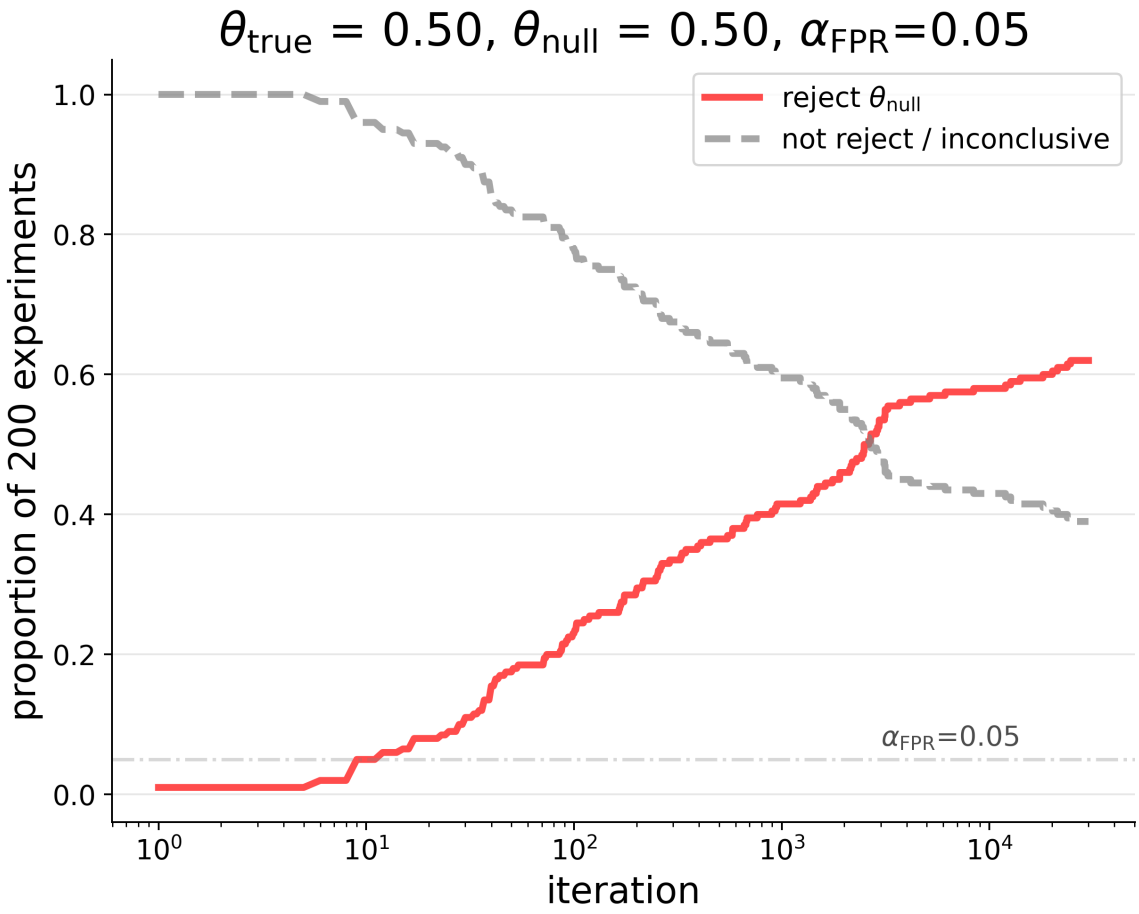


Figure 10: Cumulative decision rates across iterations for $M = 200$ fair coin experiments ($\theta_{\text{true}} = \theta_{\text{null}} = 0.5$, $\alpha_{\text{FPR}} = 0.05$, $N_{\max} = 30{,}000$), extending Figure 5 with NHST as a fourth panel; it appears in this appendix as NHST lies outside the scope of the main analysis. At each iteration N, the plotted proportions count all experiments that reached a decision at or before N; their sum is 100% at every point. *Red solid line*: reject θ_{null}. *Gray dashed line*: inconclusive. The extended horizon ($N_{\max} = 30{,}000$ vs. 1,500 in the main analysis) is needed to make the growing false-rejection trend visible; the reduced experiment count ($M = 200$ vs. 5,000) reflects the higher computational cost of evaluating p-values relative to HDI-based criteria, which also accounts for the stepped appearance of the curves.

Setting $w = \omega_{\text{goal}}$ and solving for N gives the general expected stop iteration:

$$N_{\text{goal}} \approx \frac{4z_*^2}{\omega_{\text{goal}}^2} V(\theta). \tag{7}$$

Binomial Case

For Bernoulli outcomes the posterior under the Haldane prior Beta$(0, 0)$ is Beta(s, f), where s is the number of successes and $f = N - s$ the number of failures.[23] Its exact variance is

$$\text{Var}(\theta \mid s, f) = \frac{\hat{\theta}\,(1 - \hat{\theta})}{N + 1},$$

where $\hat{\theta} = s/N$. Substituting into Equation 6 and setting $w = \omega_{\text{goal}}$ gives the exact expected stop iteration:

$$N_{\text{goal}} = \frac{4z_*^2}{\omega_{\text{goal}}^2}\,\theta(1 - \theta) - 1. \tag{8}$$

[23] This appendix uses the Haldane prior rather than the flat Beta$(1, 1)$ prior of Appendix A: the Haldane convention yields the clean -1 correction in the planning formula and is consistent with the planning function in the accompanying code (`binomial_rate_ci_width_to_sample_size` in `utils_stats.py`).

The -1 arises from the $N+1$ denominator in the exact Beta posterior variance, whereas the CLT approximation (Equation 7) treats the denominator as N and omits it. As shown in Figure 2 in Section 2.3, N_{goal} is maximised at $\theta = 0.5$, confirming that the fair coin case requires the largest sample to achieve a given precision goal, and decreases monotonically as θ moves towards 0 or 1. Equation 1 in the main text uses the CLT approximation (Equation 7), which replaces $N+1$ with N and drops the -1 term, negligible for large N.

Continuous Case

For continuous outcomes when the estimand is a mean μ with variance σ^2, substituting $V(\mu) = \sigma^2$ (estimated by sample variance s^2) into Equation 7 gives the CLT-based expected stop iteration:

$$N_{\text{goal}} \approx \frac{4z_*^2\sigma^2}{\omega_{\text{goal}}^2}. \tag{9}$$

This approximation is accurate whenever $N \geq 30$, where the Student-t critical value approaches z_*.[24]

D Precision-Based Sample Size Planning for Two-Group Settings

The HDI+ROPE framework and precision-goal approach presented in the main text for single-group Bernoulli data generalises to two-group comparisons. This appendix outlines the key adaptations for two-group binomial and continuous outcomes, together with the corresponding sample-size planning equations. For single-group continuous data, the expected stop iteration is derived in Appendix C. Empirical validation of DPitG in these two-group settings is left for future work.

Two-Group Comparisons

For two-group comparisons, whether binomial ($\theta_1 - \theta_2$) or continuous ($\mu_1 - \mu_2$), the standard error of the difference serves as the precision metric, enabling sequential two-group testing under the same HDI+ROPE decision framework (Kruschke, 2013). For the binomial case with conjugate Beta posteriors, the posterior of the difference can be evaluated by direct i.i.d. sampling from each Beta distribution separately, without Markov Chain Monte Carlo (MCMC; Gelman et al. 2013). For the continuous case, under a weakly informative prior the posterior of $\mu_1 - \mu_2$ converges to the frequentist sampling distribution by the Bernstein–von Mises theorem (Gelman et al., 2013), so the standard error from Welch's approximation provides a reliable surrogate for the posterior HDI width when $N \gtrsim 30$ (see Appendix C).

Two-Group Binomial

With independent Beta posteriors for θ_1 and θ_2, the HDI width of the difference $\theta_1 - \theta_2$ is approximated via the standard error

$$\text{SE}(\hat{\theta}_1 - \hat{\theta}_2) \approx \sqrt{\frac{\theta_1(1-\theta_1)}{N_1} + \frac{\theta_2(1-\theta_2)}{N_2}}. \tag{10}$$

Here θ_i is replaced in practice by the posterior mean $\hat{\theta}_i = s_i/N_i$; the exact Beta posterior variance replaces N_i with $N_i + 1$ (cf. the Binomial Case in Appendix C). Setting the HDI width $2z_* \,\text{SE} = \omega_{\text{goal}}$ (where z_* is the normal critical value for the chosen credible interval level; see Appendix C) and writing $N_1 = rN_2$ for group-size ratio $r = N_1/N_2 > 0$, solving for N_2 gives

$$N_2 \approx \frac{4z_*^2}{\omega_{\text{goal}}^2}\left(\frac{\theta_1(1-\theta_1)}{r} + \theta_2(1-\theta_2)\right), \quad N_1 = rN_2. \tag{11}$$

For equal group sizes ($r = 1$, $N_1 = N_2 = N$) this simplifies to

$$N \approx \frac{4z_*^2[\theta_1(1-\theta_1) + \theta_2(1-\theta_2)]}{\omega_{\text{goal}}^2}. \tag{12}$$

[24] For small samples ($N < 30$), the exact posterior for μ under a non-informative prior is a scaled Student-t with $\nu = N - 1$ degrees of freedom, giving HDI width $w = 2\,t_{\alpha/2,\,N-1}\,s/\sqrt{N}$, where $\alpha = 1 -$ CI fraction (e.g. $\alpha = 0.05$ for 95%). Setting $w = \omega_{\text{goal}}$ yields an implicit equation in N that must be solved numerically. Because precision goals are seldom met at such small N, this refinement is not recommended for sample-size planning.

Two-Group Continuous

Following Welch's approximation, the standard error of the difference of means is

$$\mathrm{SE}(\hat{\mu}_1 - \hat{\mu}_2) \approx \sqrt{\frac{\sigma_1^2}{N_1} + \frac{\sigma_2^2}{N_2}}. \tag{13}$$

Writing $N_1 = rN_2$ for group-size ratio $r = N_1/N_2 > 0$ and solving for N_2 gives

$$N_2 \approx \frac{4z_*^2}{\omega_{\text{goal}}^2}\left(\frac{\sigma_1^2}{r} + \sigma_2^2\right), \quad N_1 = rN_2, \tag{14}$$

with σ_i^2 replaced by s_i^2 when the population variances are unknown. For equal group sizes ($r = 1$, $N_1 = N_2 = N$) this simplifies to

$$N \approx \frac{4z_*^2(\sigma_1^2 + \sigma_2^2)}{\omega_{\text{goal}}^2}. \tag{15}$$

E Multiple Comparisons and Bayesian Power Analysis

Multiple Comparisons

A practical advantage of the HDI+ROPE decision framework over frequentist approaches concerns multiple comparisons. In NHST, conducting several tests simultaneously inflates the false-positive error rate (Type I), requiring corrections (e.g. Bonferroni) that reduce the power of each individual test. Frequentist confidence intervals used for simultaneous inference inherit the same dependence on the full set of intended comparisons, since their simultaneous coverage properties are tied to p-values. Consequently, both power and precision planning must account for the number of planned tests.

The Bayesian posterior is not affected in this way. Because the posterior distribution is determined solely by the observed data and the assumed data-generating process, it is unaffected by how many other comparisons were planned, unlike frequentist p-values, which depend on the full set of intended tests even when some are never conducted (Gelman et al., 2012; Kruschke, 2015). The power of any one test is therefore unaffected by how many other comparisons are planned, provided each test uses an independent model. An independent model means each comparison is fitted separately, without shared parameters or a common prior; when tests are connected through a hierarchical model, posteriors are linked across comparisons and the independence no longer holds (Gelman et al., 2012). The precision goal ω_{goal} and the stopping rule of DPitG therefore apply unchanged in a multiple-testing context, without the need for correction.

Three Types of Bayesian Power Analysis

The same design flexibility extends to how power is conceptualised in the Bayesian setting, where Kruschke (2015) distinguishes three types of power analysis that arise in the Bayesian setting.

Prospective power analysis is conducted before data collection: given a hypothesised data-generating distribution (from theory, pilot data, or prior research), one simulates experiments and asks how often the precision goal ω_{goal} is achieved within the budget $N_{\max}$. The analytical formula for N_{goal} (Equation 1) provides this as a closed-form lower bound on the required sample size; DPitG may require additional samples when the true parameter value lies near a ROPE boundary (see Section 3.2).

Retrospective power analysis uses the posterior from already-collected data as the hypothetical data generator to ask whether the completed study was sufficiently powered. In NHST this quantity is a monotone function of the p-value and therefore adds no new information beyond it (Hoenig and Heisey, 2001); in the Bayesian setting it can reveal whether the precision goal ω_{goal} was realistically achievable given the observed data, independently of the decisiveness verdict.

Replication power asks: if the experiment were repeated exactly, what is the probability of achieving the goal in the new study, informed by the results of the first? In the DPitG framework this is straightforward to compute by using the initial posterior as the prior for the follow-up and re-running the stopping rule; NHST has no principled access to a posterior-derived data generator and cannot support this form of analysis (Kruschke, 2015).

F Regarding Bayes Factors

This appendix concerns the use of the Bayes Factor (BF) as a *sequential stopping rule*; this role is distinct from its use in Bayes Factor Design Analysis (BFDA), which is a prospective planning tool discussed in Section 4.

The BF is a widely used Bayesian measure for comparing two hypotheses. For a null hypothesis H_0 and an alternative H_1, the Bayes Factor in favour of the null is defined as the ratio of marginal likelihoods:

$$\mathrm{BF}_{01} = \frac{p(\text{data} \mid H_0)}{p(\text{data} \mid H_1)}. \tag{16}$$

A value $\mathrm{BF}_{01} > 1$ constitutes evidence in favour of H_0; a value $\mathrm{BF}_{01} < 1$ constitutes evidence against it. The scale proposed by Jeffreys (1961) and refined by Kass and Raftery (1995) provides conventional thresholds: $\mathrm{BF}_{01} > 3$ is typically taken as "substantial" evidence for the null, and $\mathrm{BF}_{01} < 1/3$ as substantial evidence against it.

For the Bernoulli setting with z successes in N trials, Kruschke (2015) (Chapter 13.3.2) uses a point null $H_0 : \theta = \theta_{\text{null}}$ and a uniform alternative $H_1 : \theta \sim \mathrm{Beta}(1, 1)$. The marginal likelihood under H_0 is simply $\theta_{\text{null}}^{z}(1-\theta_{\text{null}})^{N-z}$, and under H_1 it is the Beta–Binomial integral

$$p(\text{data} \mid H_1) = \int_0^1 \theta^z (1-\theta)^{N-z}\, d\theta = B(z+1,\, N-z+1), \tag{17}$$

where $B(\cdot, \cdot)$ denotes the Beta function, giving the closed-form expression

$$\mathrm{BF}_{01} = \frac{\theta_{\text{null}}^{z}(1-\theta_{\text{null}})^{N-z}}{B(z+1,\, N-z+1)}. \tag{18}$$

The sequential stopping rule is: continue sampling until either $\mathrm{BF}_{01} \geq 3$ (accept H_0) or $\mathrm{BF}_{01} \leq 1/3$ (reject H_0), or the budget $N_{\max}$ is exhausted.

Sequential Behaviour

Unlike frequentist p-values, BF stopping is valid under optional stopping: because the Bayes Factor depends only on the observed data and the assumed model, not on the sampling intention or the number of interim looks; sequential monitoring does not inflate it (Rouder, 2014). The question is therefore not whether BF stopping is theoretically principled, but whether it achieves adequate operating characteristics in practice.

Kruschke (2015) evaluates those characteristics across $M = 1{,}000$ simulated sequences of up to $N_{\max} = 1{,}500$ flips, comparing BF stopping against NHST, the HDI+ROPE algorithm, and PitG (his Figures 13.6 and 13.7). The main findings relevant to this paper are as follows.

When the null is true ($\theta_{\text{true}} = \theta_{\text{null}} = 0.5$)

This is the setting shown in Kruschke's Figure 13.6 (for which Figure 5 of this paper is analogous). The BF stopping rule converges to an asymptotic false rejection rate (i.e, the *false alarm rate*) of just over 20%, lower than NHST, which diverges to 100%, but substantially higher than PitG, which maintains a 0% false rejection rate (Kruschke, 2015). As demonstrated in this paper, DPitG also achieves a 0% false rejection rate. The BF correctly accepts the null for the remaining sequences, although it can do so on the basis of precise or imprecise data indiscriminately.

When the null is false ($\theta_{\text{true}} = 0.65,\ \theta_{\text{null}} = 0.5$)

This setting is displayed in Kruschke's Figure 13.7 (which is captured in the middle panel of Figure 6). The BF falsely accepts the null in approximately 40% of sequences despite the true parameter lying outside the ROPE, an alarmingly higher false acceptance rate, whereas the precision-based methods yield 0 false acceptances.

Parameter estimation at the stopping point

Because the BF stopping rule fires on the *extremeness* of the likelihood ratio rather than the *width* of the posterior, it can accept or reject with very little data. As the three lower rows of Figures 13.6 and 13.7 of Kruschke (2015) illustrate, $\hat{\theta}$ at stopping is noticeably biased away from θ_{true} in both the accept and the reject case, more so than the HDI+ROPE algorithm, and substantially more than PitG. This is the fundamental problem identified in Kruschke (2015): a stopping rule based on extremeness in the data automatically biases the sample toward extreme estimates; only a precision-based stopping rule guarantees near-unbiased parameter estimation.

Key Limitations in Context

Three structural limitations follow from the BF's definition.

No effect-size criterion

BF_{01} compares the point null $\theta = \theta_{\text{null}}$ against the prior-averaged alternative. While the alternative prior implicitly encodes a distribution over effect sizes, the BF provides no explicit mechanism to require the posterior to fall cleanly inside or outside a practically meaningful interval: a parameter value just outside the ROPE is treated identically to one that is far from it. The HDI+ROPE decision framework addresses this explicitly by requiring the HDI to fall cleanly inside or outside the ROPE before a verdict is issued.

Sensitivity to the alternative prior

BF_{01} depends on the choice of prior $p(\theta \mid H_1)$ (Equation 17). Kruschke's choice of a uniform Beta$(1, 1)$ prior is one convention; different priors yield different numerical BF values and therefore different stopping points, even for identical data. The HDI+ROPE stopping rule has no such ambiguity: its behaviour depends only on the posterior and the pre-specified ROPE and ω_{goal}.

No precision guarantee

The BF can stop early on imprecise posteriors. In contrast, PitG and DPitG require the HDI width to fall below ω_{goal} before any verdict is issued, providing a structural guarantee that accepted or rejected conclusions rest on adequately precise estimates.

These limitations notwithstanding, BF stopping and HDI+ROPE are not mutually exclusive. As noted in Section 4, a design requiring both a conclusive HDI+ROPE verdict *and* a minimum Bayes Factor is a natural extension of DPitG, combining the precision guarantee of the latter with the evidential strength measure of the former.

G Practical Design Guidance and Broader Context

Setting the ROPE

Three design inputs must be fixed before data collection: the ROPE, the precision goal ω_{goal}, and the maximum sample size $N_{\max}$. The precision goal should satisfy $\omega_{\text{goal}} \leq \Delta_{\text{ROPE}}$ and directly determines the expected stop iteration N_{goal} (Equation 1); $N_{\max}$ is set by budget. Of these three subjective inputs the ROPE is the most consequential. It should be determined from domain knowledge rather than from the data, a requirement that forces the researcher to commit to a practically meaningful effect-size threshold before observing outcomes (Kruschke, 2018). In clinical research the Minimal Clinically Important Difference (MCID) provides a principled basis (Jaeschke et al., 1989); in bioequivalence testing, regulatory guidelines define the equivalence margin (Schuirmann, 1987); in industrial quality control, engineering tolerances provide an equivalent principled basis. When no domain standard exists, sensitivity analyses across a range of ROPE widths are recommended before committing to a single value (Lakens et al., 2018).

Broader Adoption Considerations

Three considerations support broader adoption of the DPitG framework.

First, formally accepting the null via the HDI+ROPE framework's criterion parallels the objective of frequentist equivalence testing through the Two One-Sided Tests (TOST) procedure (Schuirmann, 1987; Lakens et al., 2018), providing a Bayesian counterpart to a methodology established in regulatory and applied settings (U.S. Food and Drug Administration, 2010).

Second, because ω_{goal}, the ROPE, and $N_{\max}$ must all be specified before data collection begins, DPitG is structurally compatible with pre-registration standards (Nosek et al., 2018), directly reducing the researcher degrees of freedom, the analytic choices that enable post-hoc optimisation of results, that drive selective reporting.

Third, the replication crisis has been partly attributed to underpowered studies whose imprecise estimates fail to replicate (Open Science Collaboration, 2015; Gelman and Carlin, 2014); DPitG addresses this at the design stage by making precision a non-negotiable stopping criterion rather than an aspiration, one that is structurally enforced by construction.

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